



数据可视化基础

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课程安排

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第2周	第3讲：感知和认知	第4讲：数据可视化基础（1）
第3周	第5讲：数据可视化基础（2）	实验1：可视化图表研究
第4周	第6讲：可视化生成方法（1）	实验2：制作网页（1）
第5周	第7讲：可视化生成方法（2）	实验2：制作网页（2）
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第7周	第9讲：跨媒体数据可视化	实验4：可视化流程复现（1）
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第11周	第13讲：高维数据可视化	实验5：可视化创作（1）
第12周	第14讲：层次数据可视化	实验5：可视化创作（2）
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第15周	第16讲：交互	实验5：可视化创作（5）
第16周	实验5展示	实验5展示

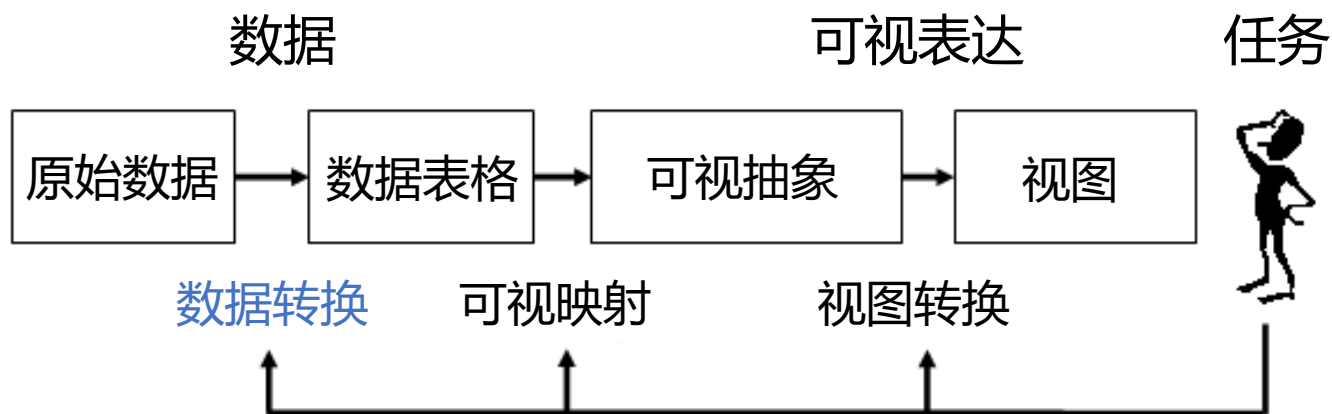
目录

- 数据转换
- 基础图表

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- 数据转换
- 基础图表

可视化模型



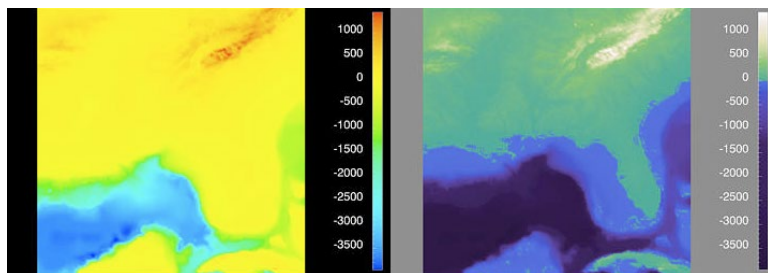
JAVA可视化工具包Prefuse中使用的可视化参考模型

数据转换

- 标准化 (Normalization)
- 采样 (Sampling)
- 离散化 (Discretization)
- 拟合 (Fitting)
- 聚类 (Clustering)
- 降维 (Dimensionality reduction)

标准化

- 定义：将数据从原始范围转换到特定范围或分布
- 作用：对不同数据统一处理
- 例子：绩点换算
- 可视化中的应用：颜色映射，位置映射



标准化

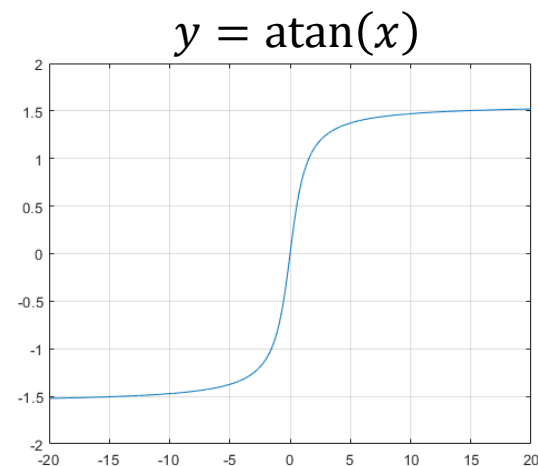
- 线性转换至[0, 1]

$$x' = \frac{x - \min}{\max - \min}$$

- 反正切转换至[-1, 1]

$$x' = \text{atan}(x) * \frac{2}{\pi}$$

- 自定义转换



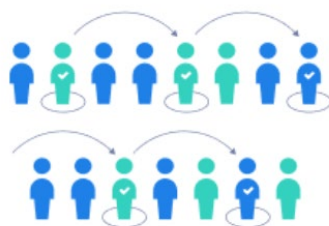
采样

- 定义：从目标总体中抽取一部分个体作为样本
- 作用：减轻数据量大带来的负担（计算、渲染...）
- 例子：标记重捕法
- 可视化中的应用：大规模数据绘制

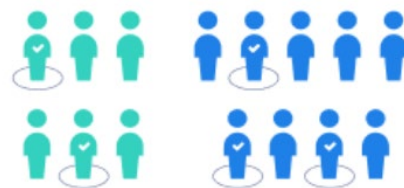
采样



随机采样
(Random Sampling)

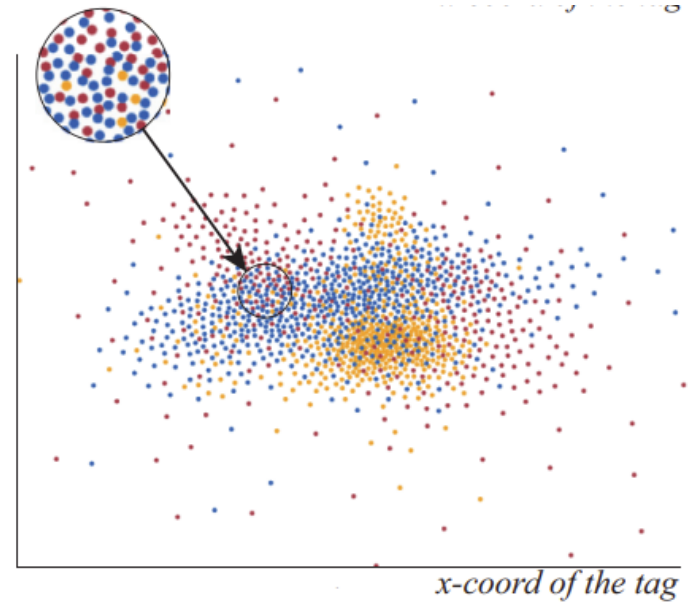
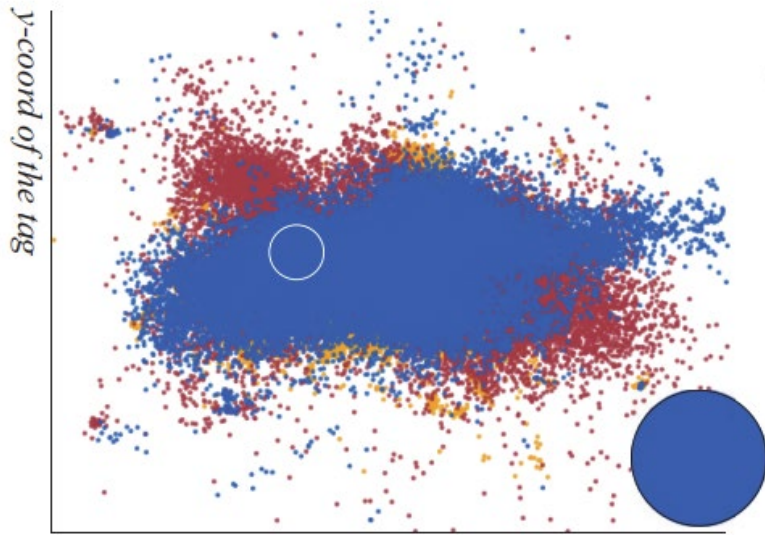


系统采样
(Systematic Sampling)



分层采样
(Stratified Sampling)

采样



Chen et al., Visual Abstraction and Exploration of Multi-class Scatterplots, IEEE TVCG 14'

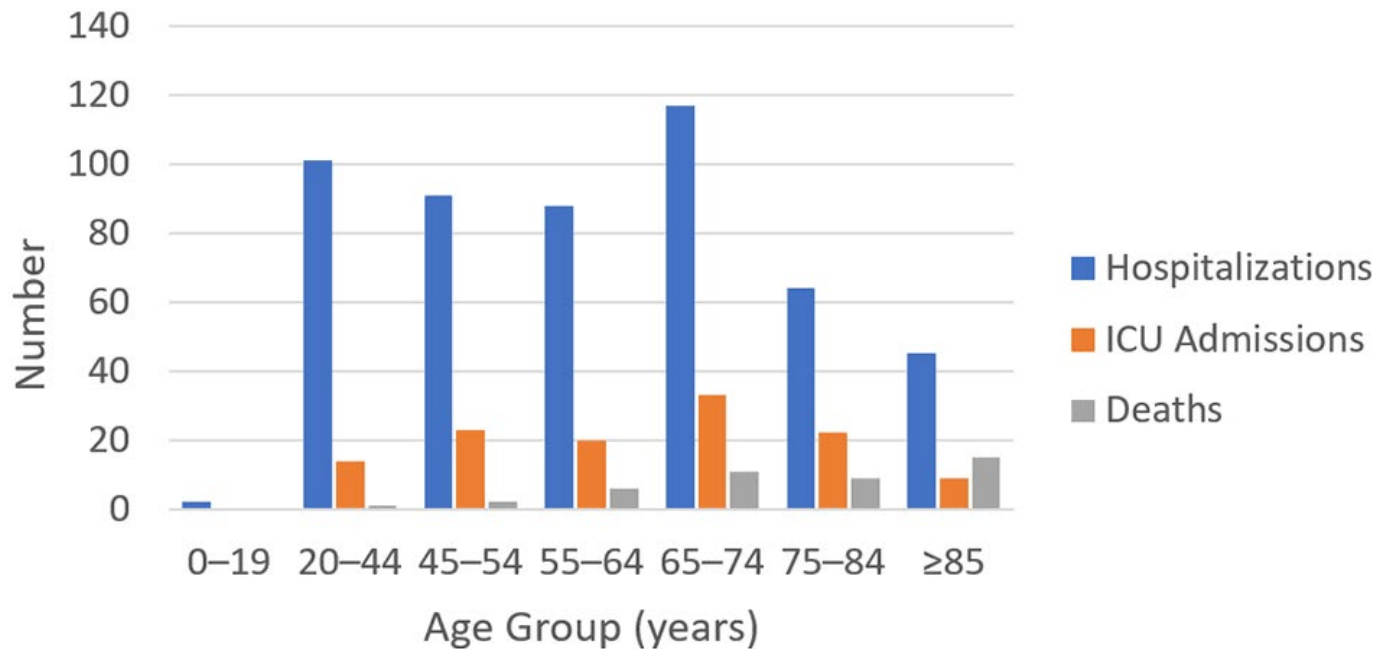
离散化

- 定义：将连续值划分成区间
- 作用：将无限空间转化为有限空间
- 例子：按年龄分组
- 可视化中的例子：为离散型数据选取颜色编码



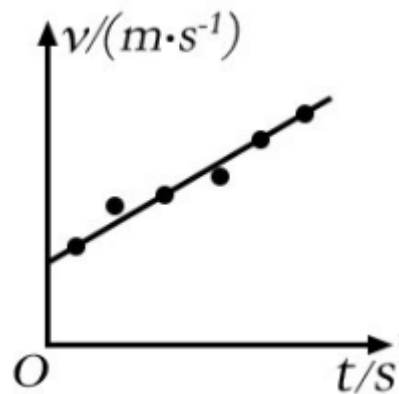
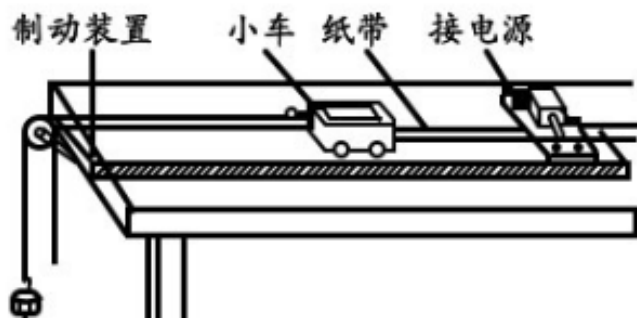
离散化

- 确定组距
 - 考虑数据分布和语义



拟合

- 定义：选择适当曲线来概括被观测数据
- 作用：解析变量之间的关系、推测未被记录的数据
- 例子：计算加速度
- 可视化中的例子：拟合概率分布曲线



拟合

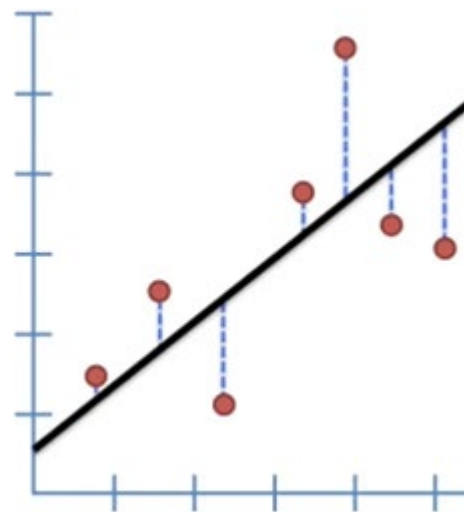
- 思路：距离平均差距最小
- 方法

- 线性回归，非线性回归

$$\min_y \sum_{i=1}^n (y'_i - y_i)^2$$

- 偏最小二乘回归 (PLSR)

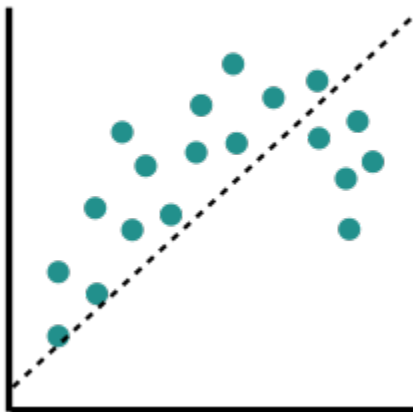
$$\min_{\vec{y}} \sum_{i=1}^n (\vec{y}'_i - \vec{y}_i)^2$$



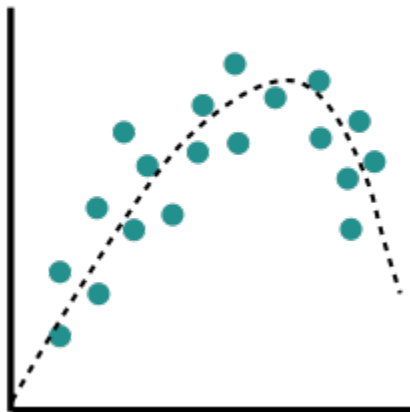
拟合

- 找到合适的拟合程度

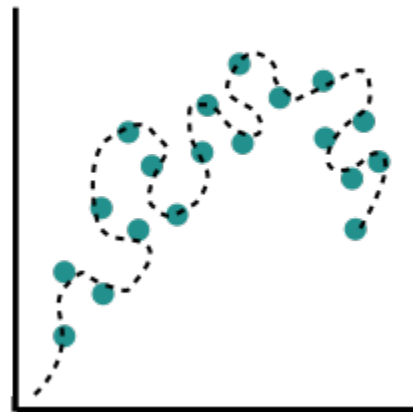
欠拟合



恰当的拟合

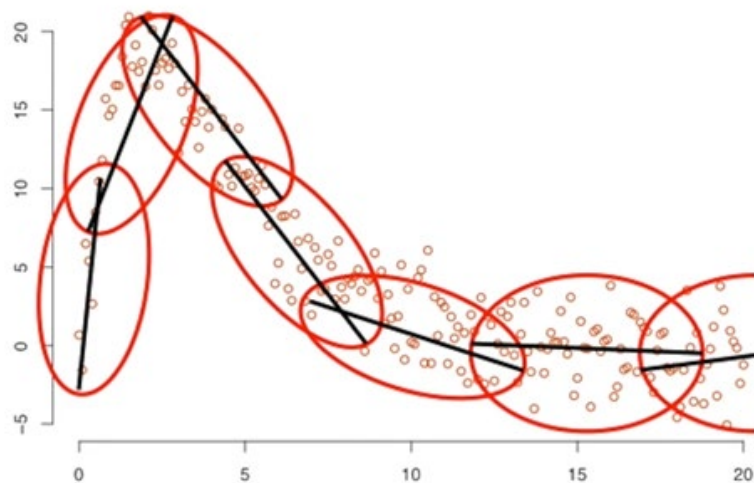


过拟合

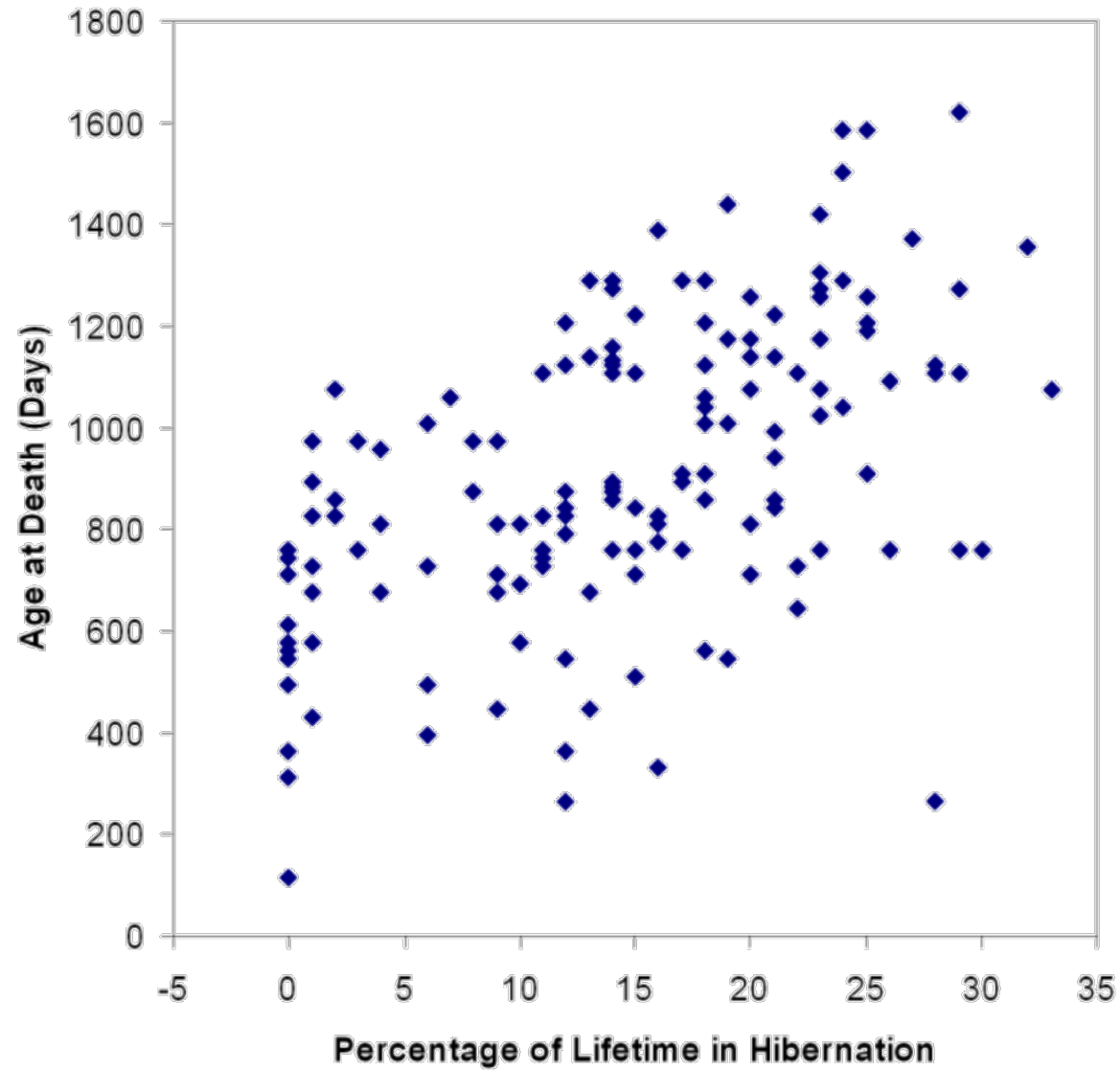


拟合

- 局部多项式回归拟合 (LOESS)

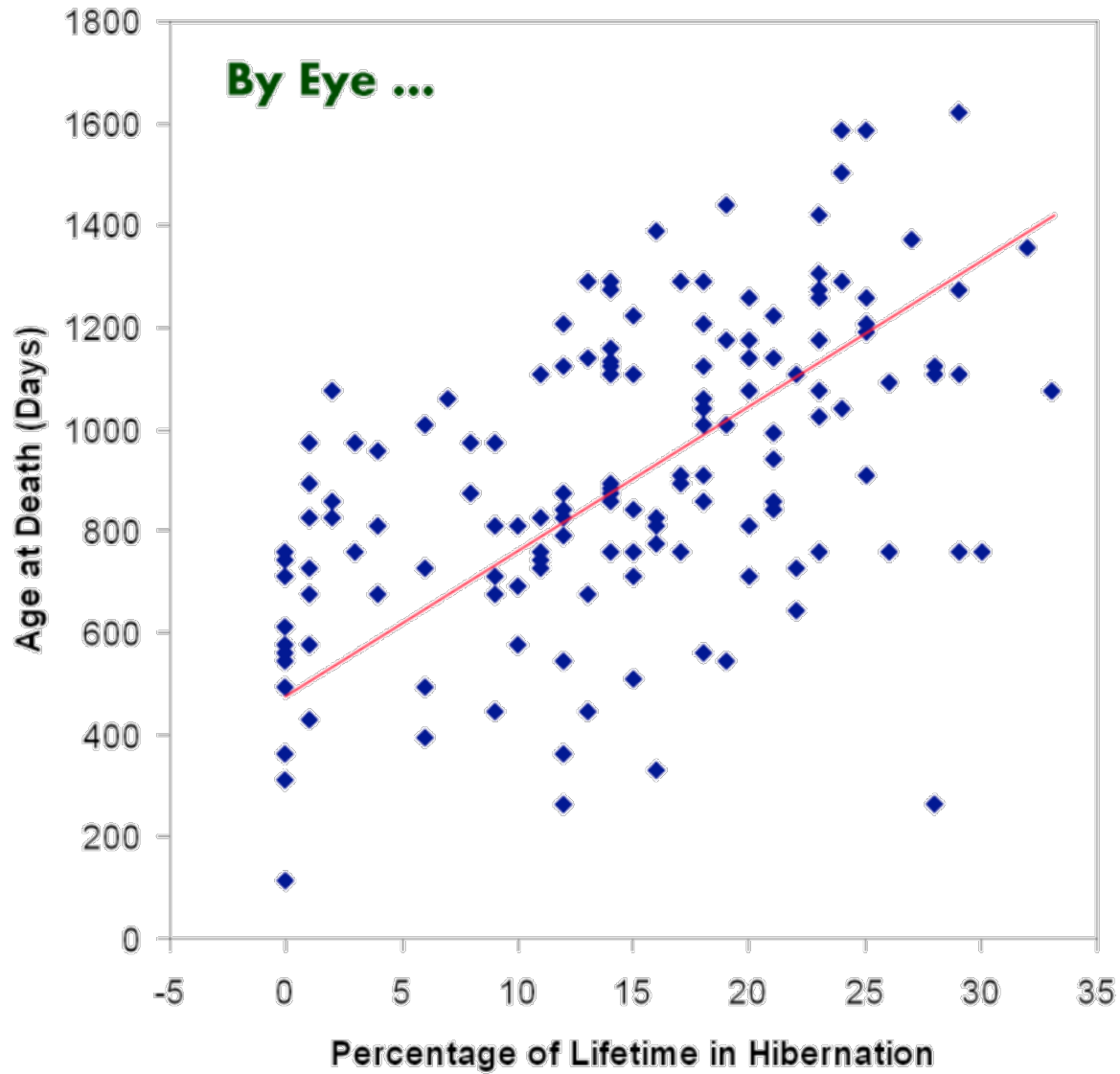


拟合



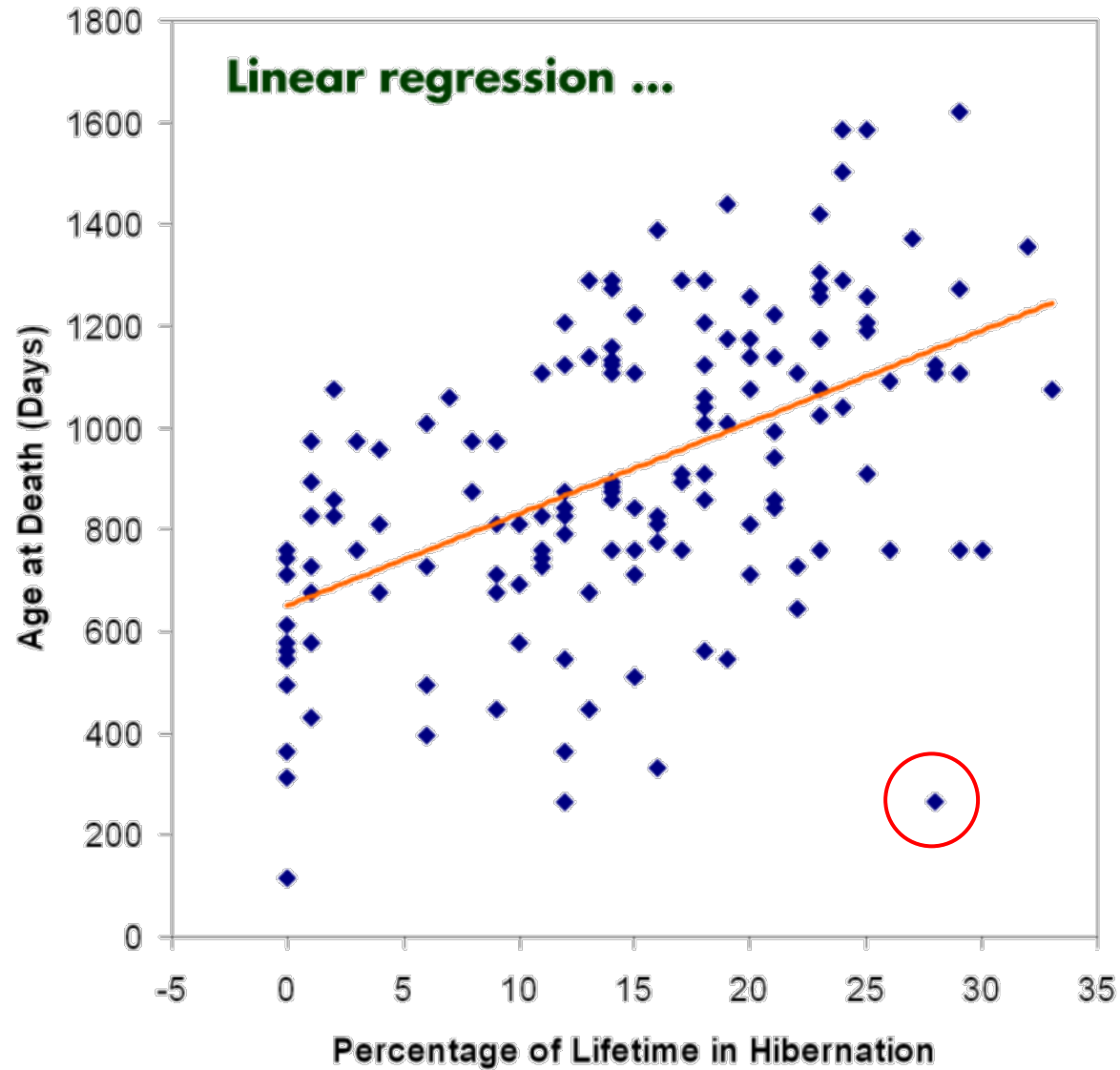
[The Elements of Graphing Data. Cleveland 94]

拟合



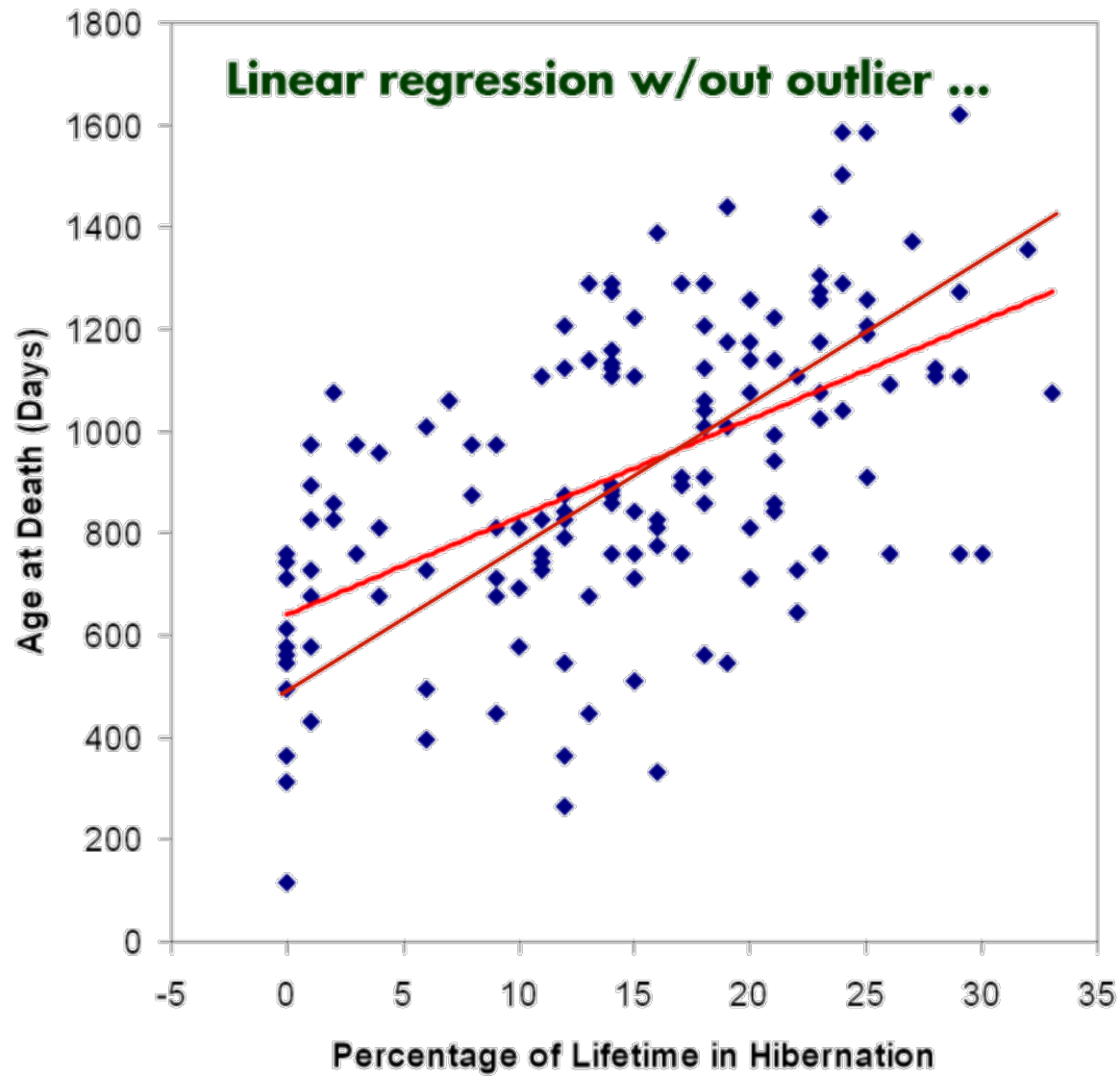
[The Elements of Graphing Data. Cleveland 94]

拟合



[The Elements of Graphing Data. Cleveland 94]

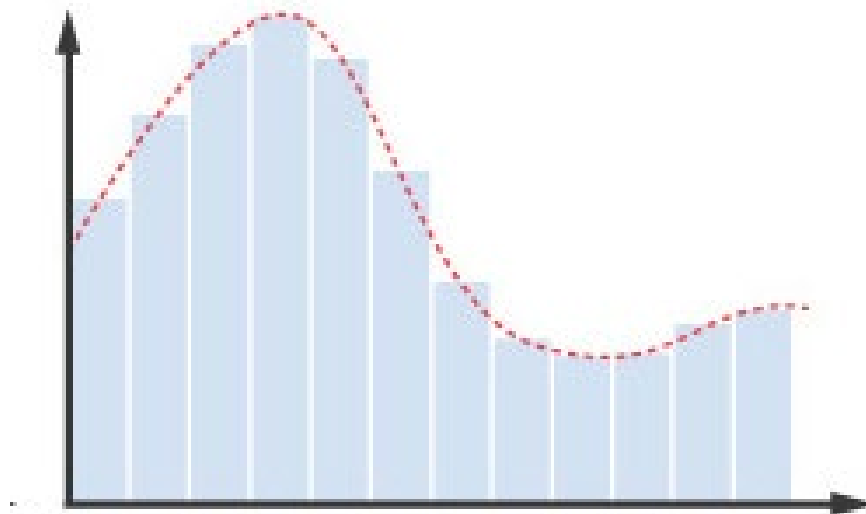
拟合



[The Elements of Graphing Data. Cleveland 94]

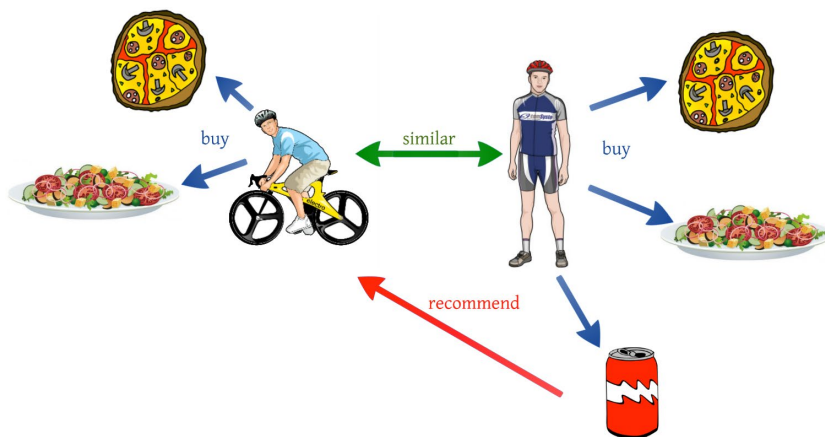
拟合

- 概率分布曲线



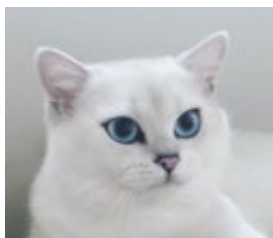
聚类

- 定义：按相似度对个体分组
- 作用：总结特征，挖掘模式...
- 例子：分析用户偏好，定向推荐
- 可视化中的例子：为类别型属性挑选颜色



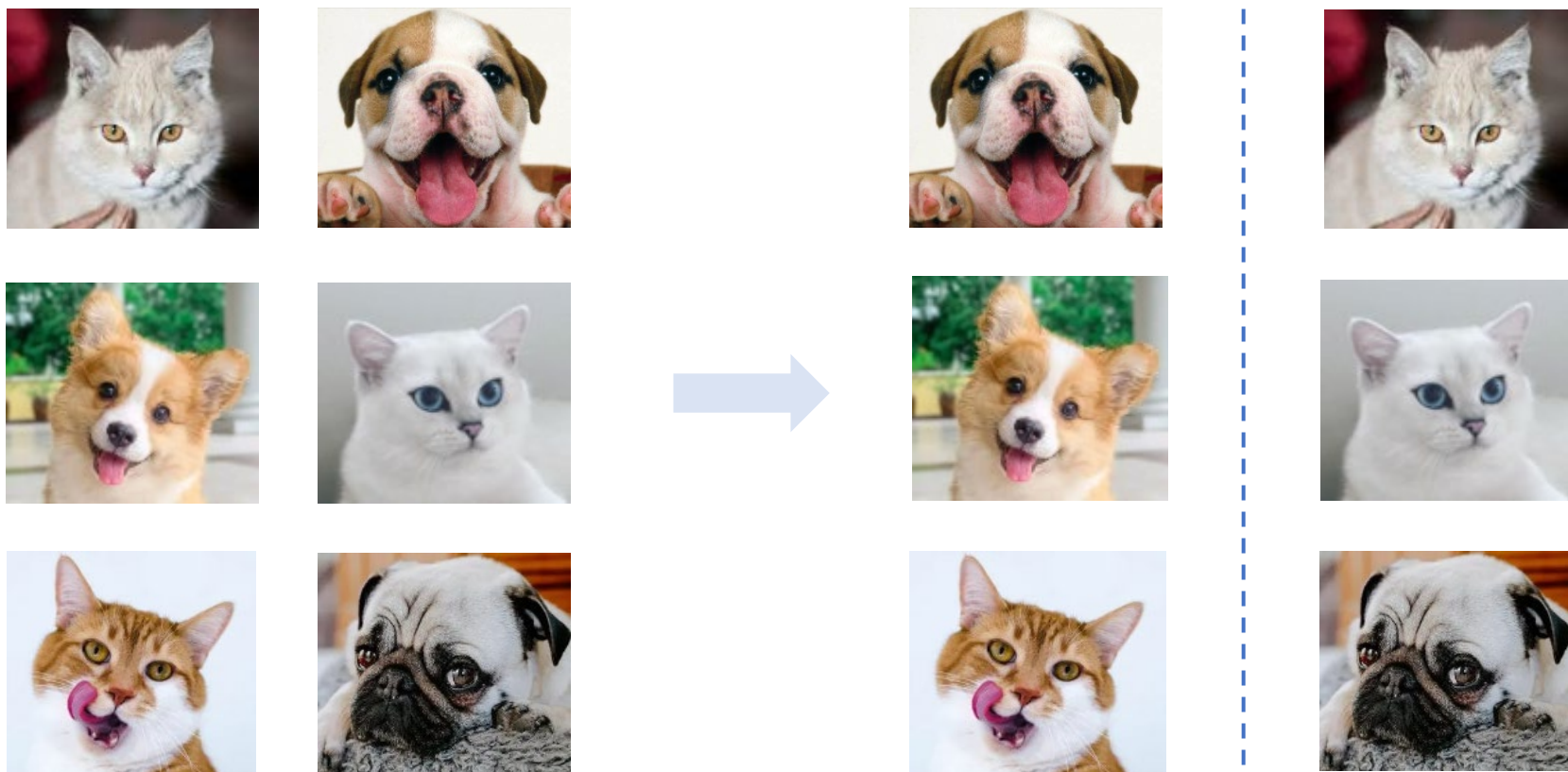
聚类vs分类

- 分类：有预定义的类别



聚类vs分类

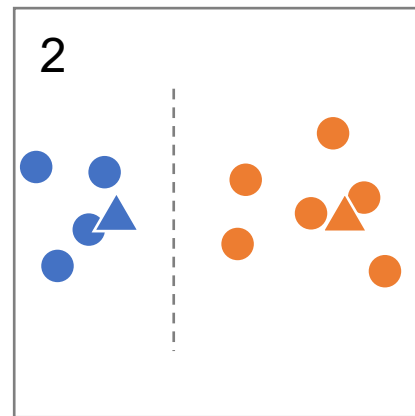
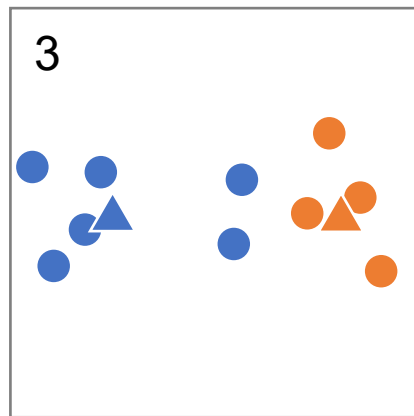
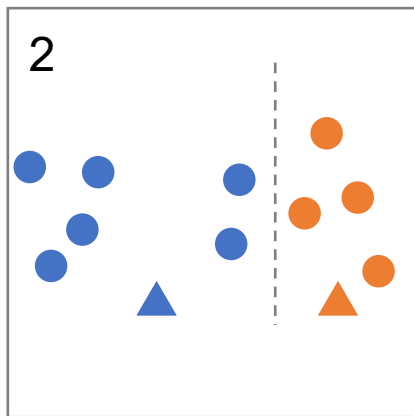
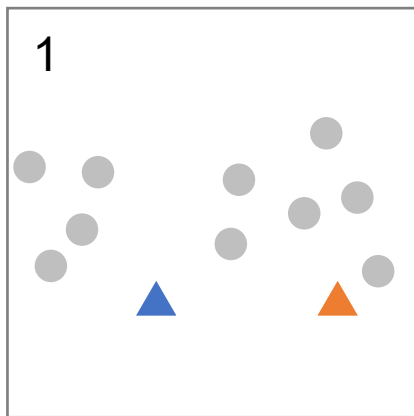
- 聚类：没有预定义类别，根据相似度分组



聚类

- k -means

1. 在数据空间随机生成 k 个种子
2. 每条数据记录和最相似的种子聚类
3. 更新种子为聚类中心
4. 重复2-3至聚类结果稳定



聚类

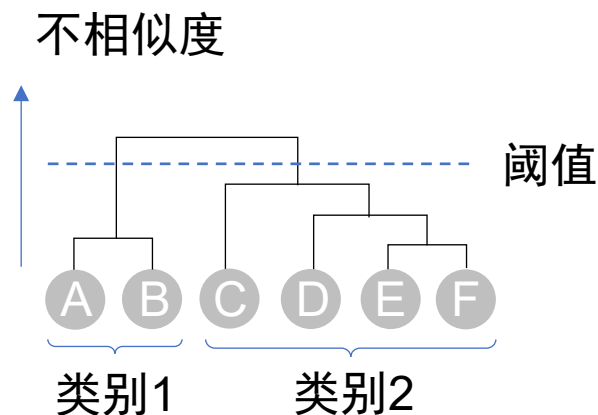
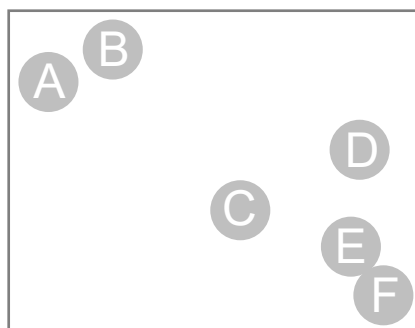


<http://tech.nitoyon.com/en/blog/2013/11/07/k-means/>

聚类

- 层次聚类

1. 计算任意两条记录之间的相似度
2. 根据相似度大小依次将记录连在一起，生成树形结构
3. 根据用户指定的聚类数量或不相似度阈值确定聚类结果



聚类→代表语义的颜色

Input term	Top clustered images	Canonical color
taxi		
lizard		
saffron		

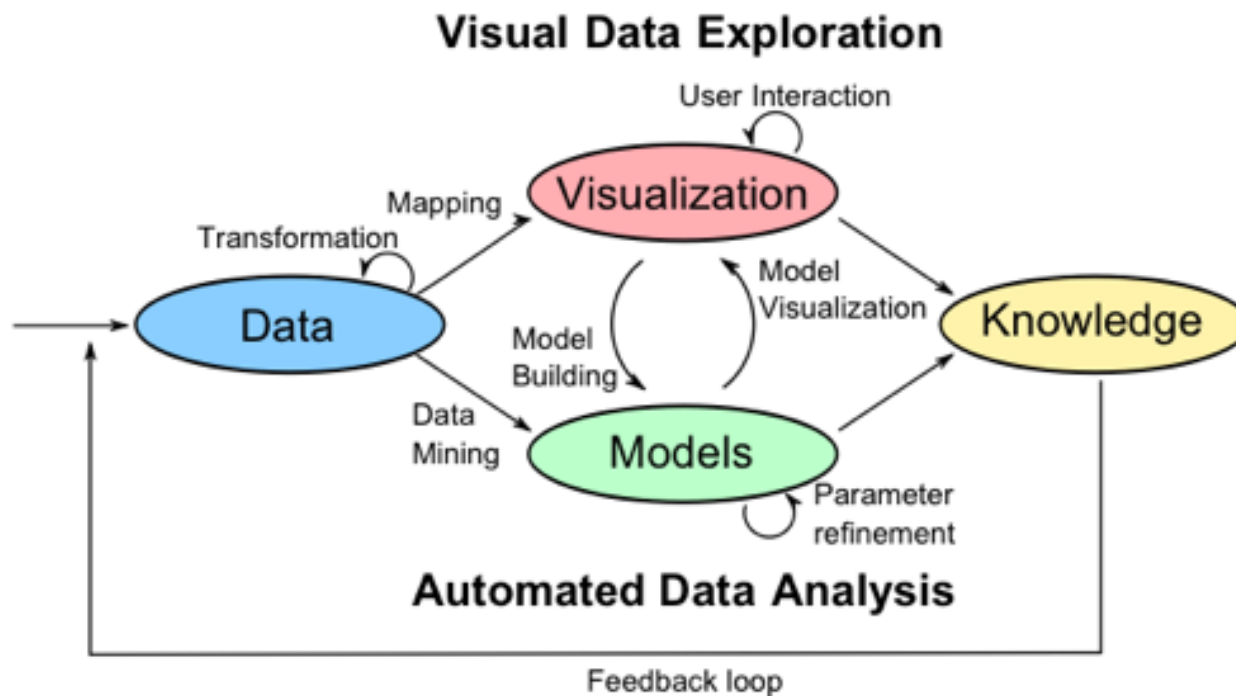
关键词搜图

使用层次聚类提取图片中的相似颜色

Setlur et al., A Linguistic Approach to Categorical Color Assignment for Data Visualization, IEEE TVCG' 16

可视分析

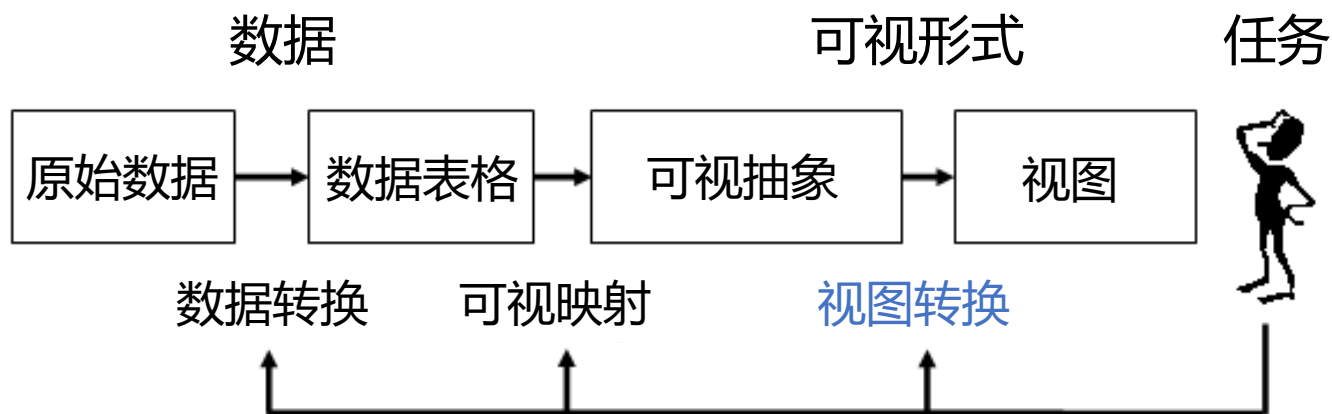
- 基于交互界面，人和机器合作分析数据



目录

- 数据转换
- 基础图表

可视化模型



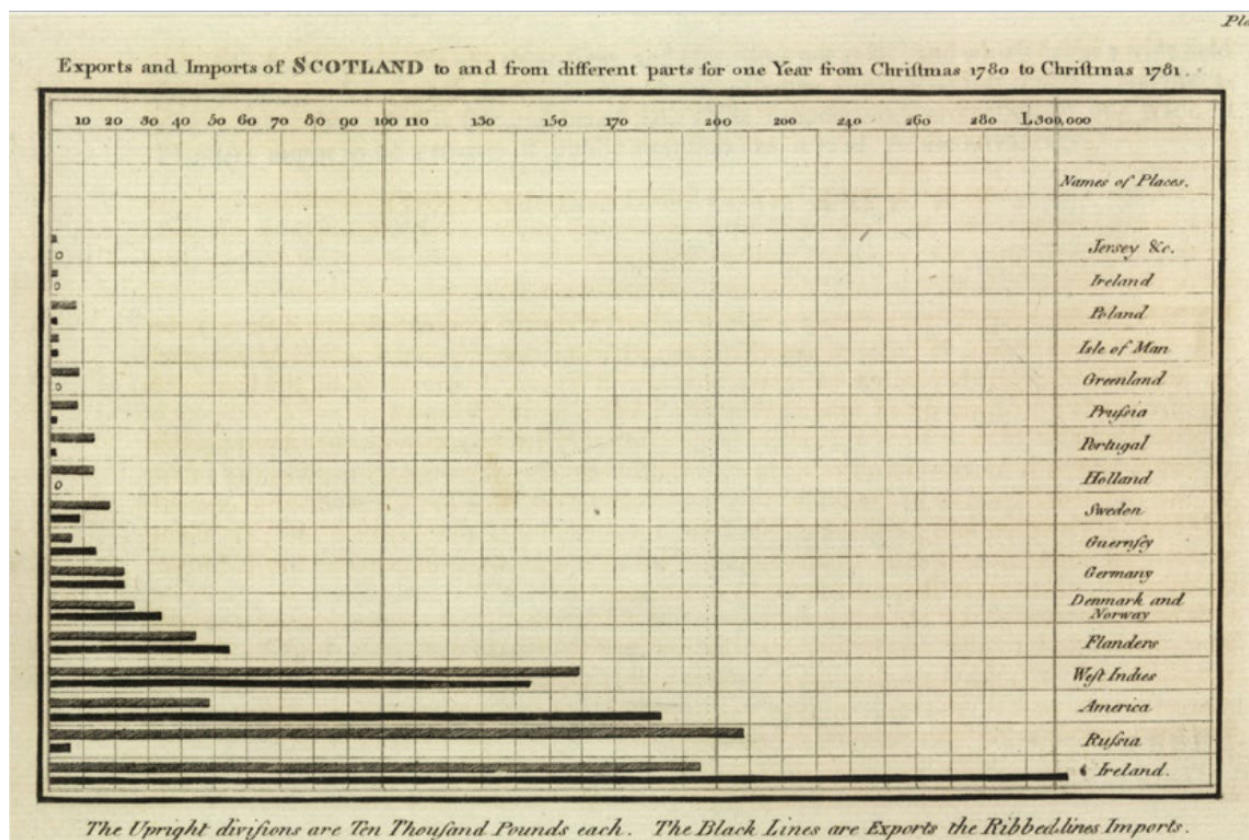
JAVA可视化工具包Prefuse中使用的可视化参考模型

基础图表

- 柱形图 (bar chart)
- 折线图 (line chart)
- 饼图 (pie chart)
- 散点图 (scatter plot)
- 盒须图 (box plot)

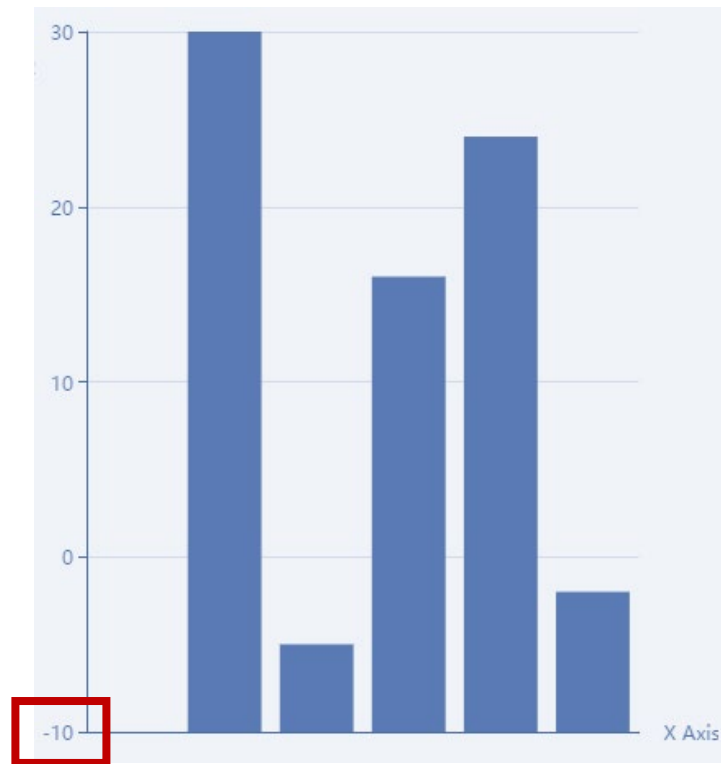
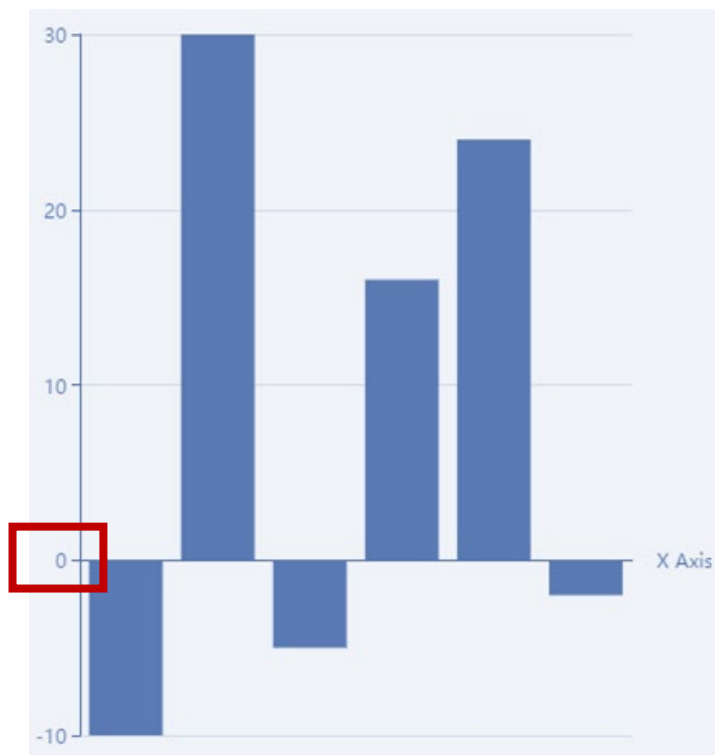
柱形图

- 用长方形的高度或长度展示数值

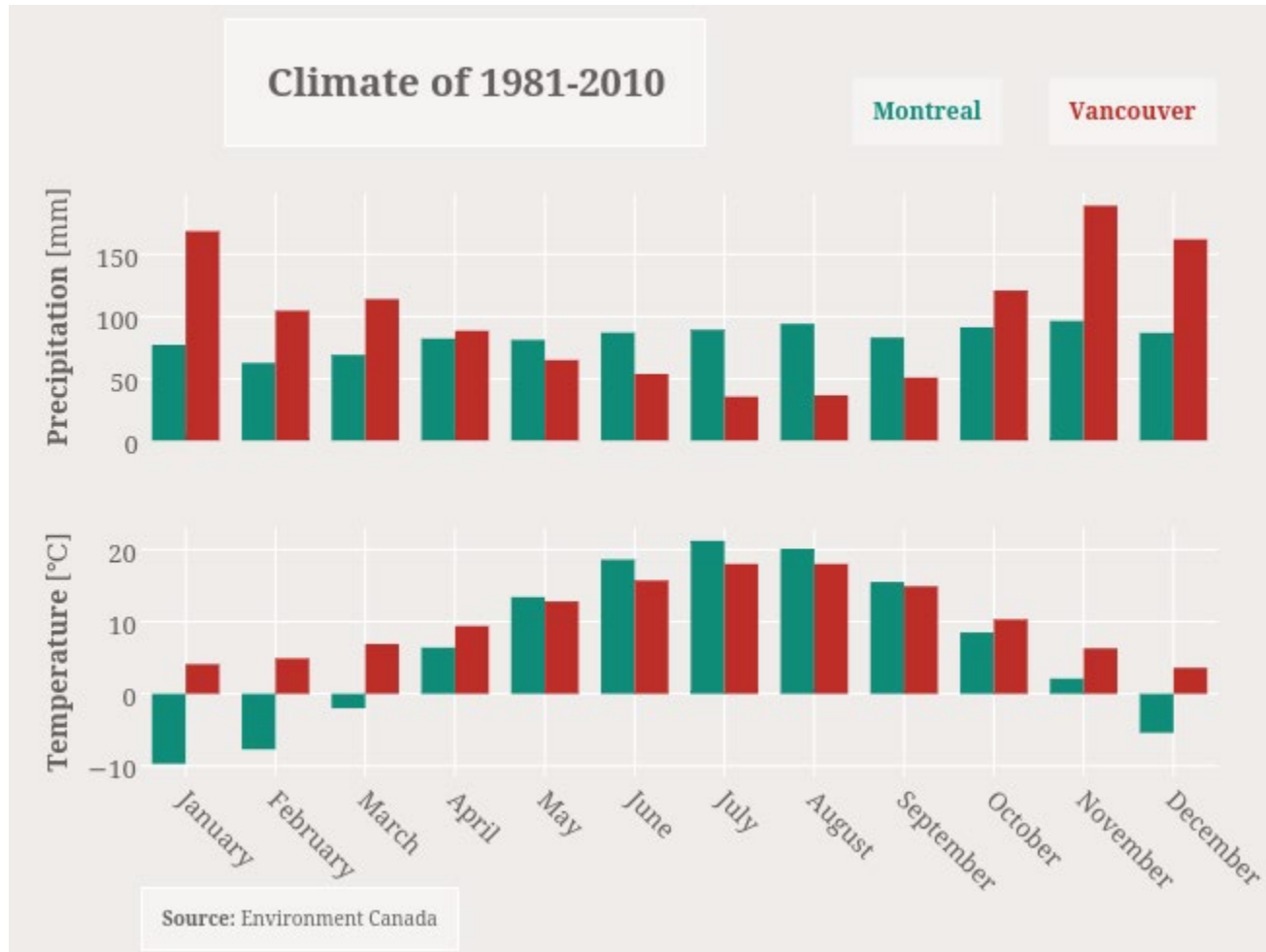


数值基线

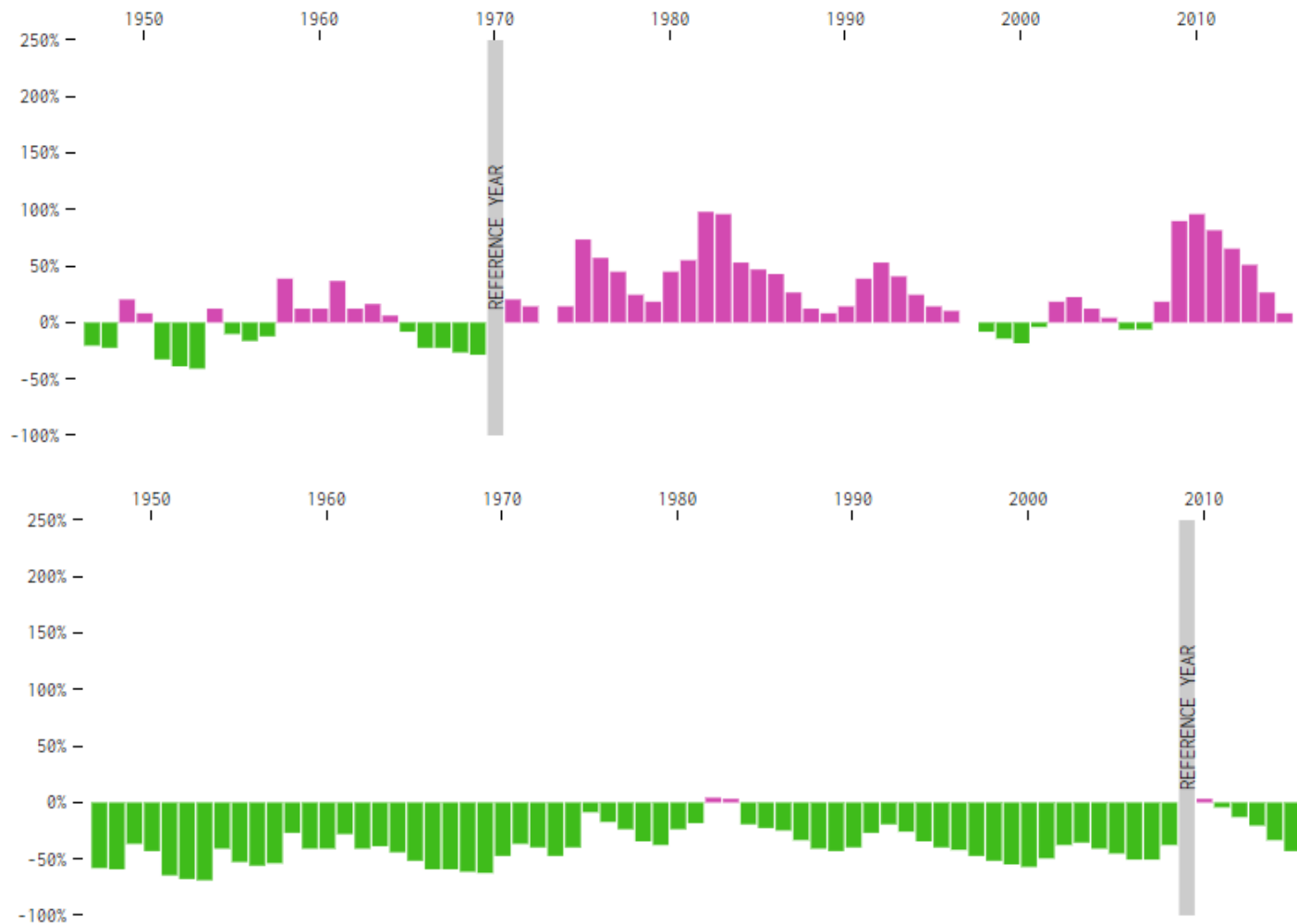
- 负数数值怎么画？



数值基线

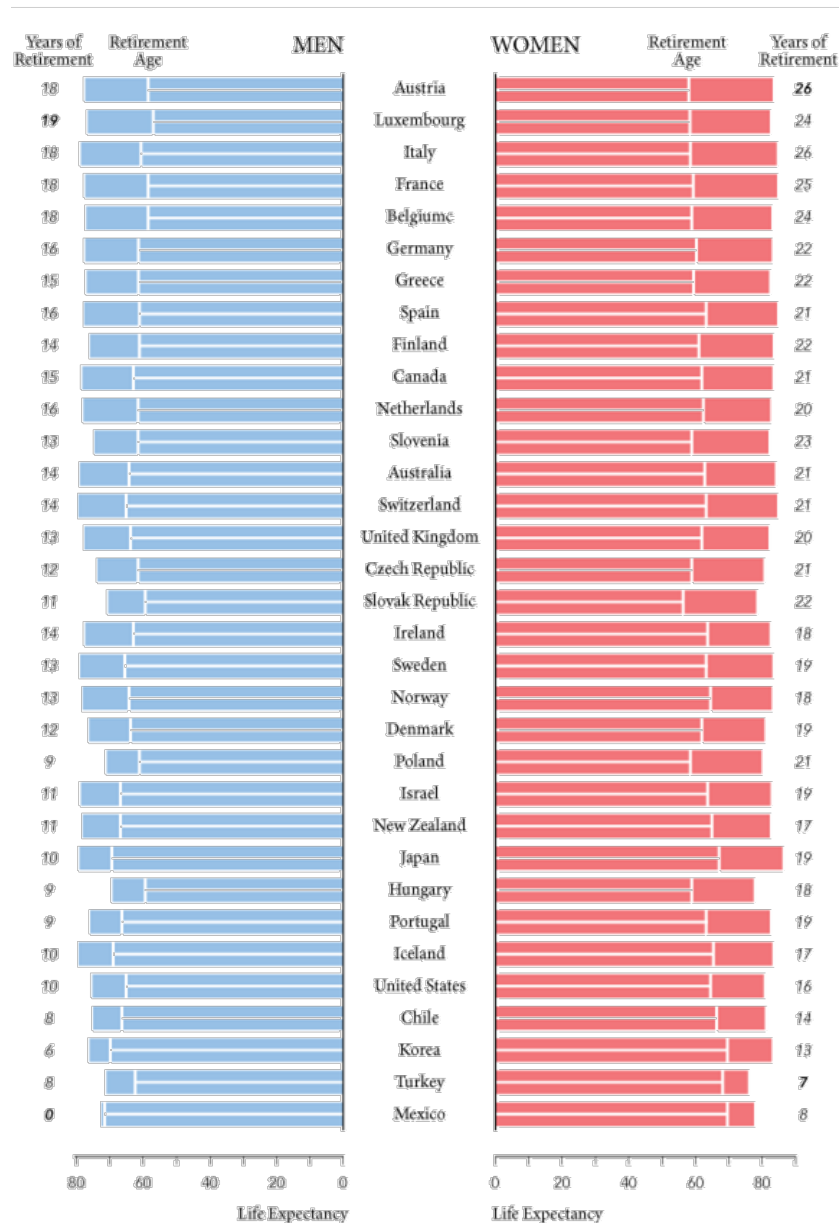


数值基线



<https://flowingdata.com/2016/11/23/shift-your-point-of-view/>

柱形图变种

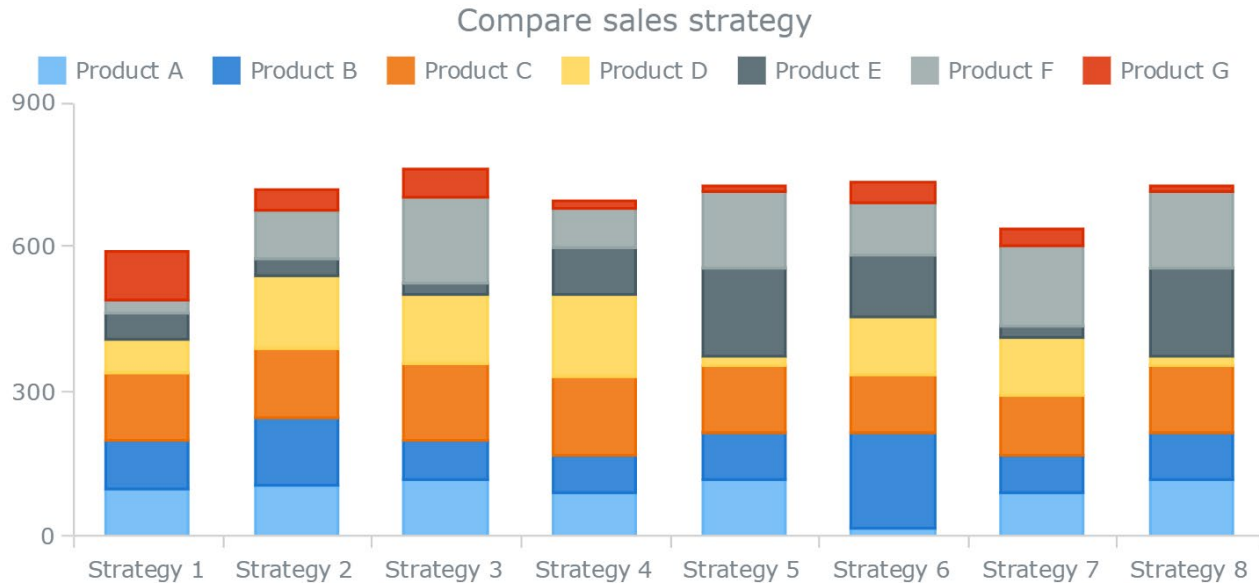


柱形图变种



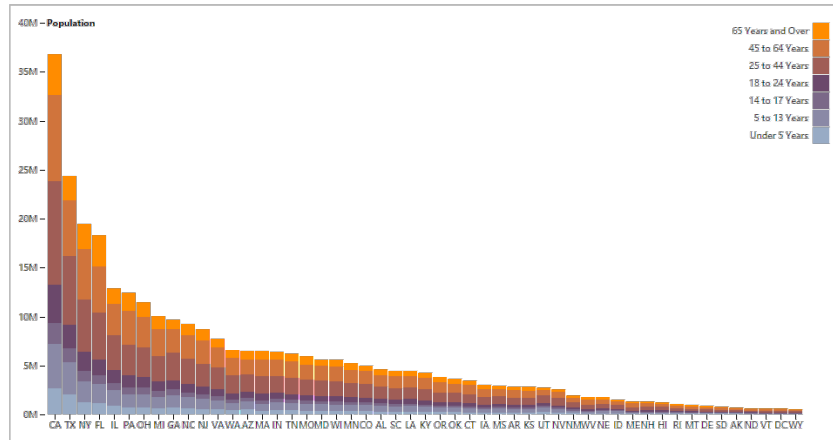
柱形图变种

- 堆叠柱形图
 - 将同类的柱子堆叠起来
 - 可用于比较部分和整体

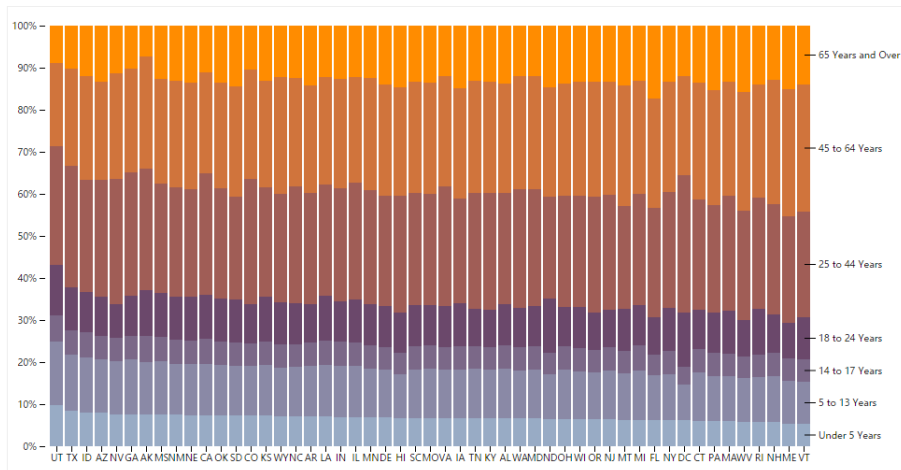


柱形图变种

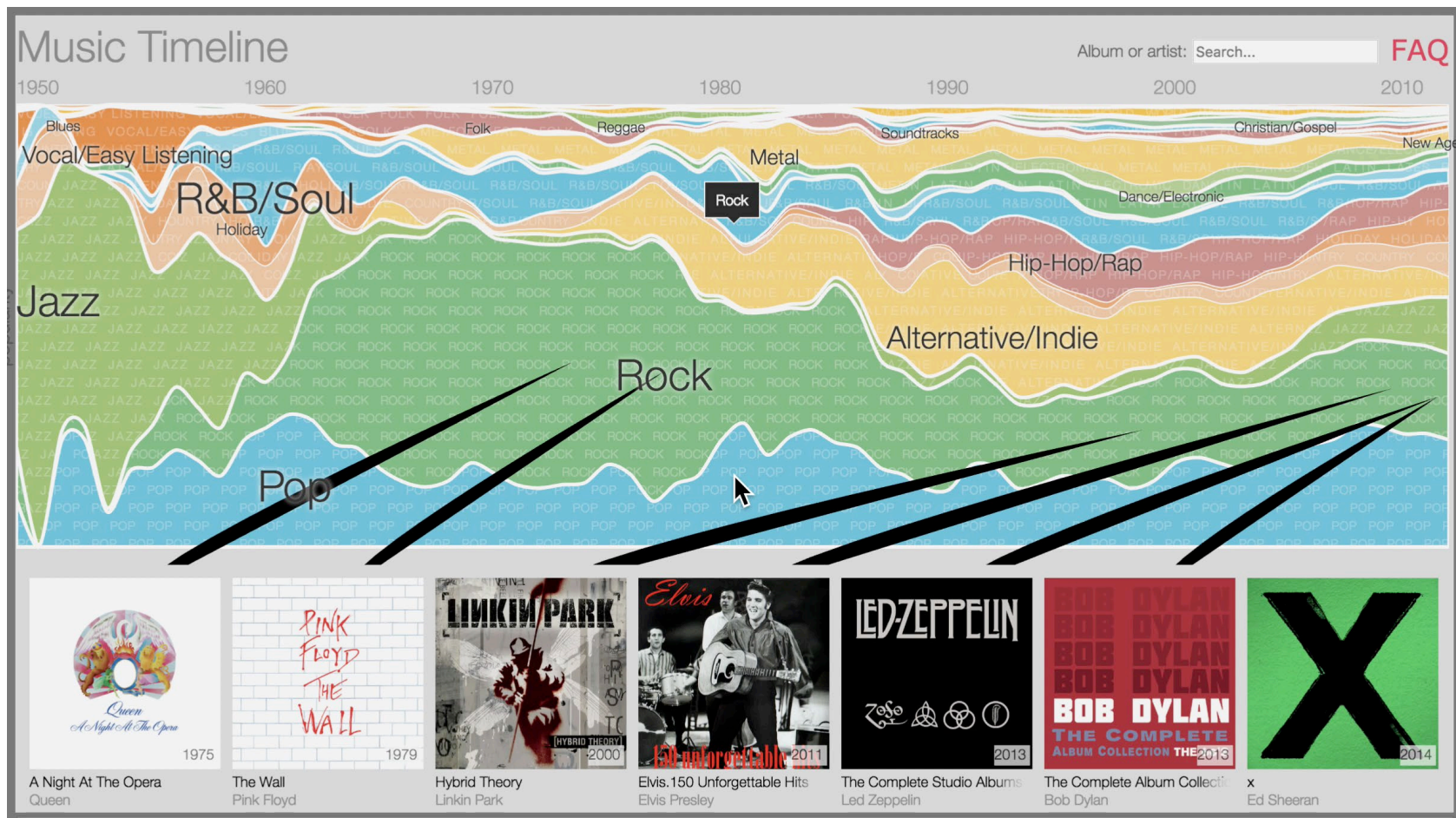
Stacked Bar Chart



Normalized Stacked Bar Chart

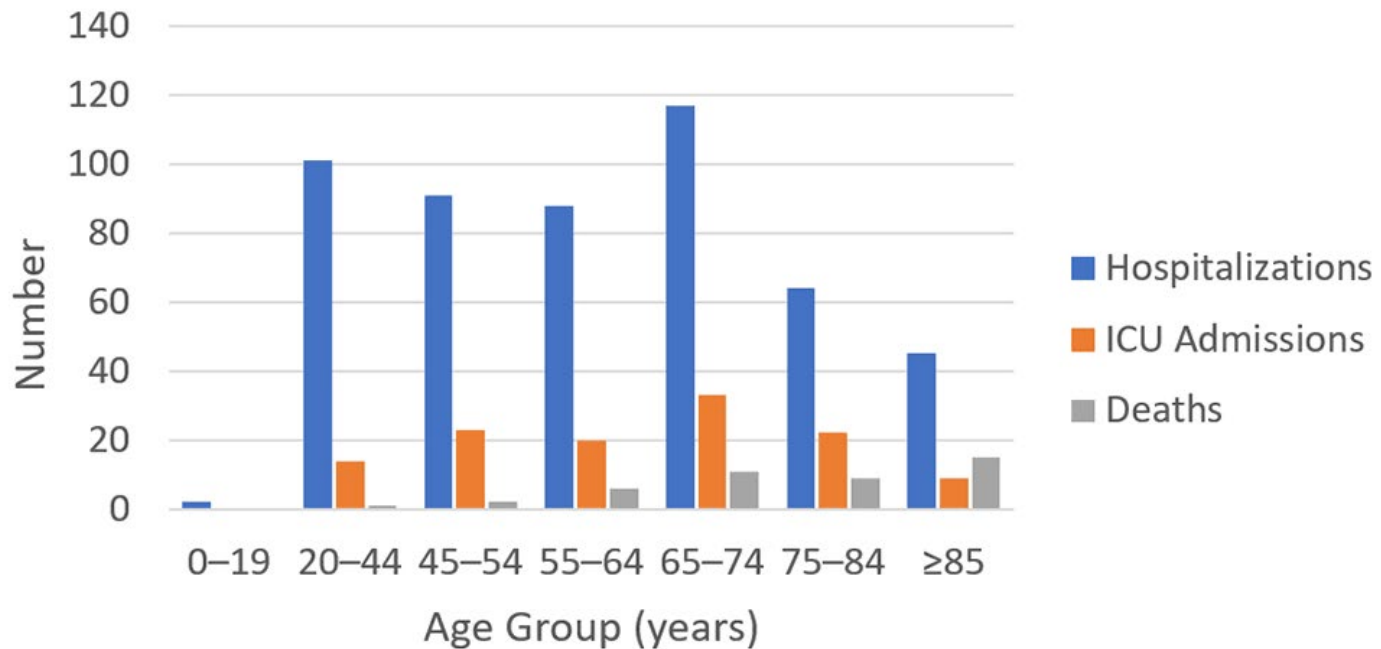


堆叠图



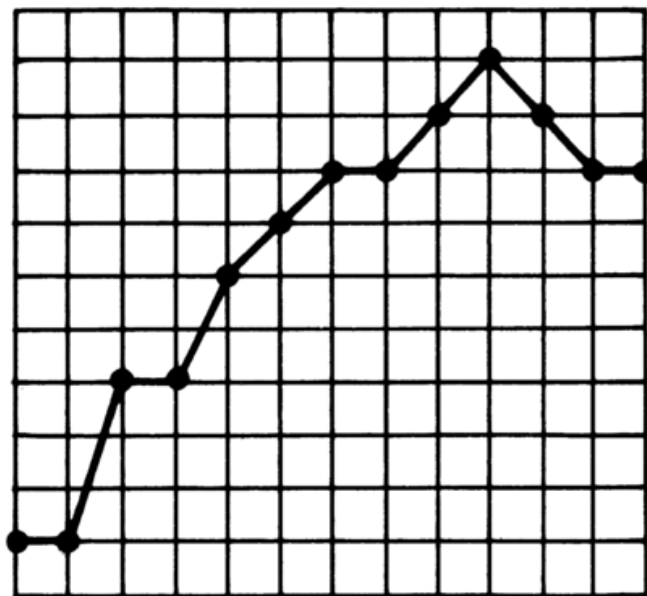
柱形图特例

- 直方图 (Histogram)
 - 确定组距后统计数量



折线图

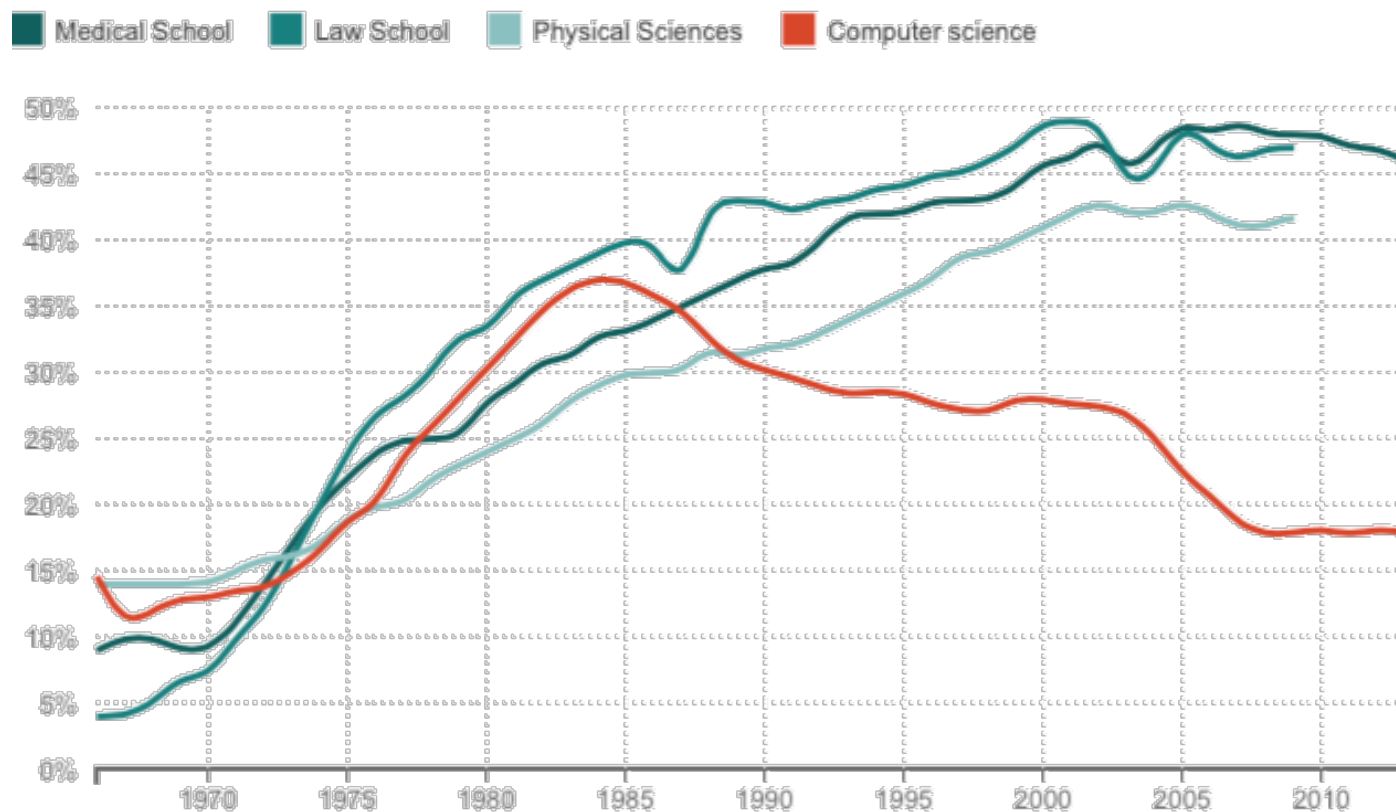
- 通过将一系列数据点连成线的方式表示信息
- 擅长展示趋势



折线图

What Happened To Women In Computer Science?

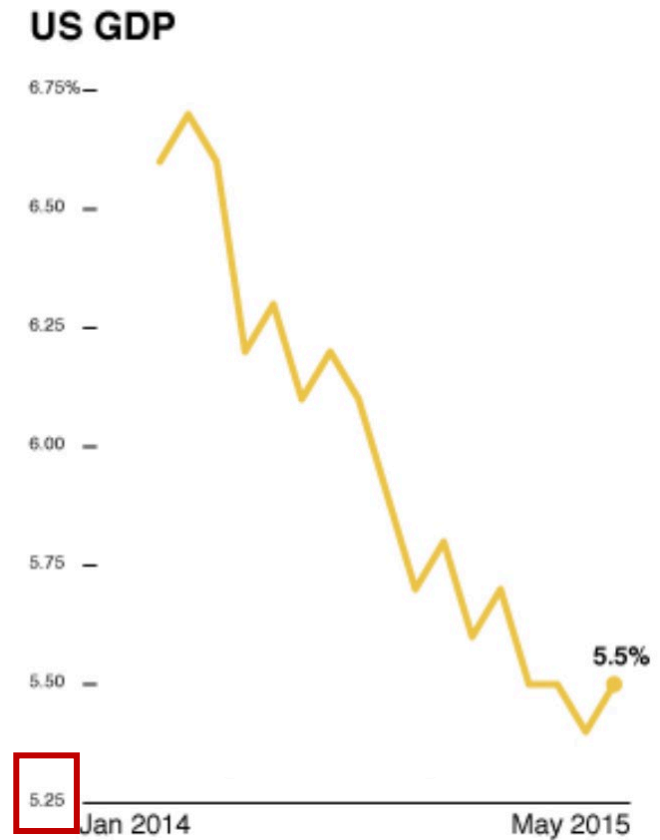
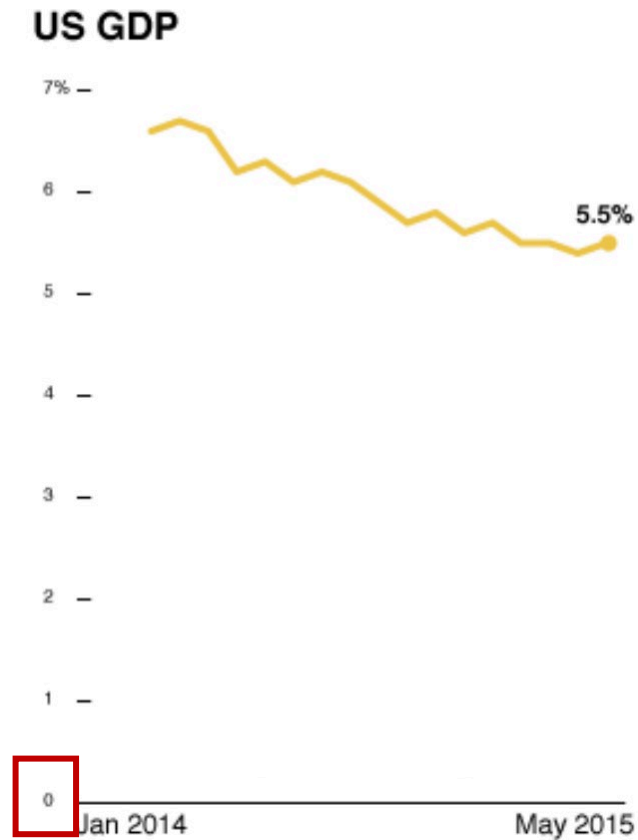
% Of Women Majors, By Field



Source: National Science Foundation, American Bar Association, American Association of Medical Colleges

Credit: Quechung Bu/NPR

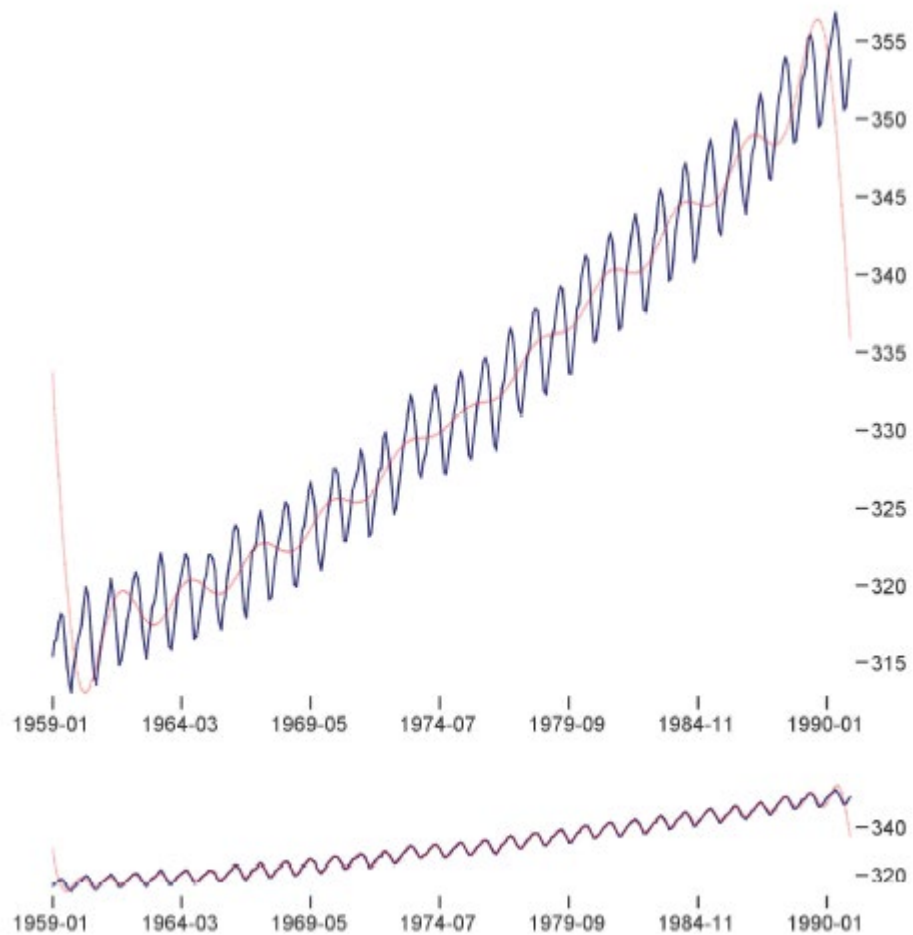
数值基线



横纵比



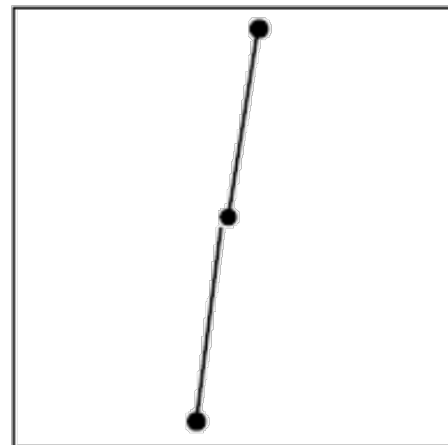
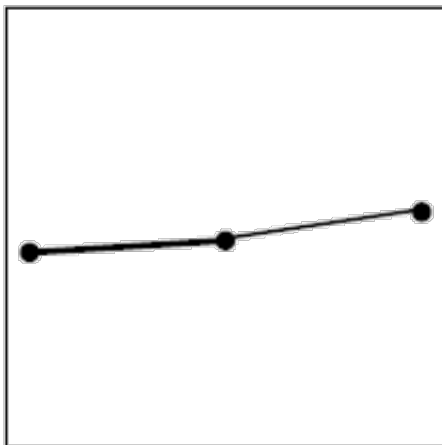
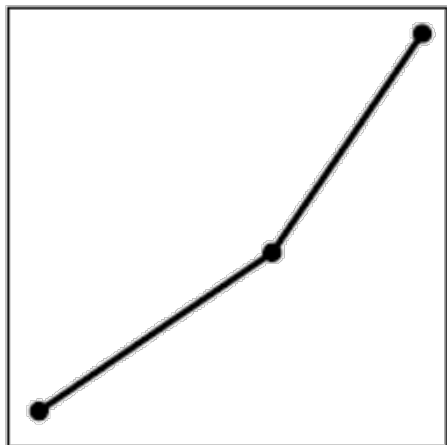
横纵比



二氧化碳浓度变化

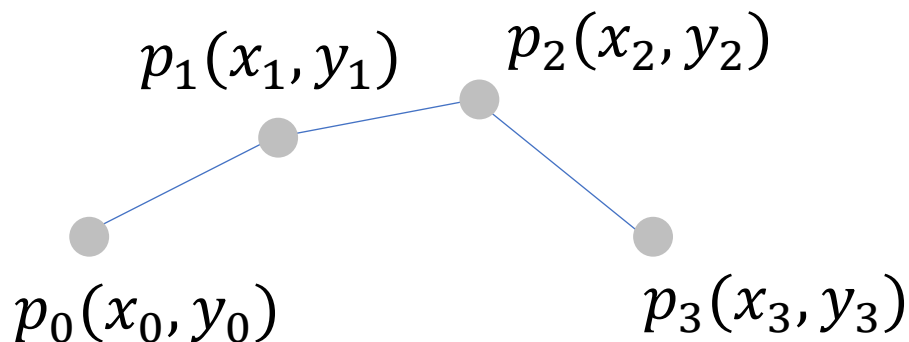
横纵比

- 坡度接近 45° 观感更好



横纵比

- 坡度优化



$$s_i = \frac{y_i - y_{i-1}}{x_i - x_{i-1}} \quad s = \frac{y_{\max} - y_{\min}}{x_{\max} - x_{\min}}$$

$$\alpha = \frac{\text{width}}{\text{height}} = ?$$

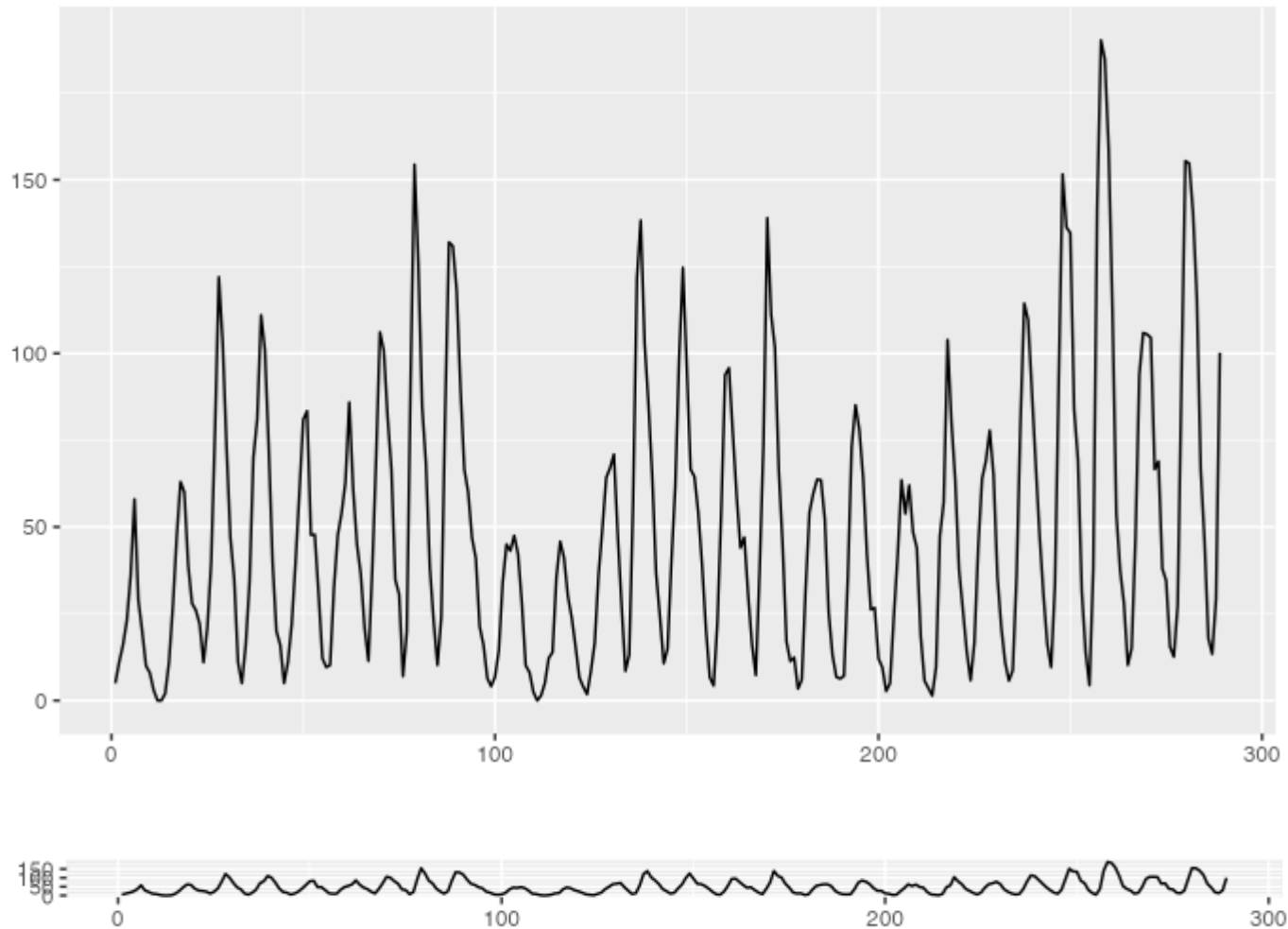
$$\text{median}\left(\left|\frac{s_i}{\alpha}\right|\right) = 1$$

$$\alpha = \frac{\text{median}(|s_i|)}{s}$$

$$\text{mean}\left(\left|\frac{s_i}{\alpha}\right|\right) = 1$$

$$\text{mean}(\text{atan}\left(\left|\frac{s_i}{\alpha}\right|\right)) = 45^\circ$$

效果



https://jrnold.github.io/ggthemes/reference/bank_slopes.html

折线图变种

- Sparklines
 - 小型隐藏轴折线图
 - 可嵌入文本或成组显示

Using d3.js, we can fairly easily draw SVG-based sparklines. This is 2013 historical stock prices for

Google  **\$1084.75**. And this is for **Facebook**  **\$55.57**. And this is for

Apple  **\$550.77**. Each sparkline has 244 data points, but it's condensed very nicely.

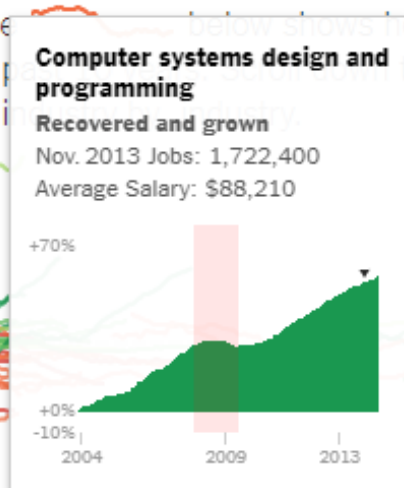
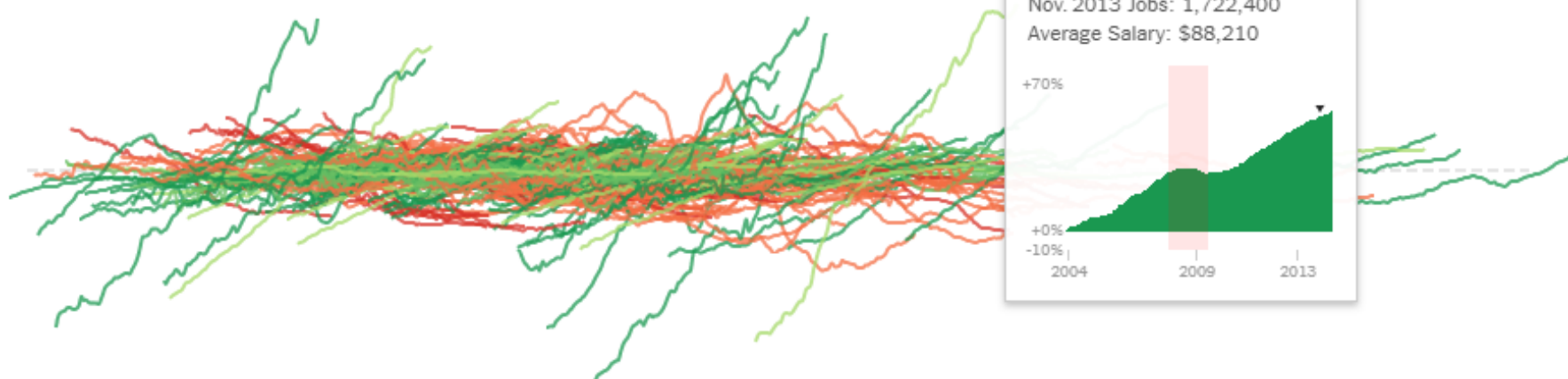
	A	B	C	D	E	F	G	H	I
1									
2	5	6	4	8	2	7			
3	2	5	7	8	9	0			
4	9	4	6	3	5	1			

折线图变种

How the Recession Reshaped the Economy, in 255 Charts

By JEREMY ASHKENAS and ALICIA PARLAPIANO Updated: JUNE 6, 2014

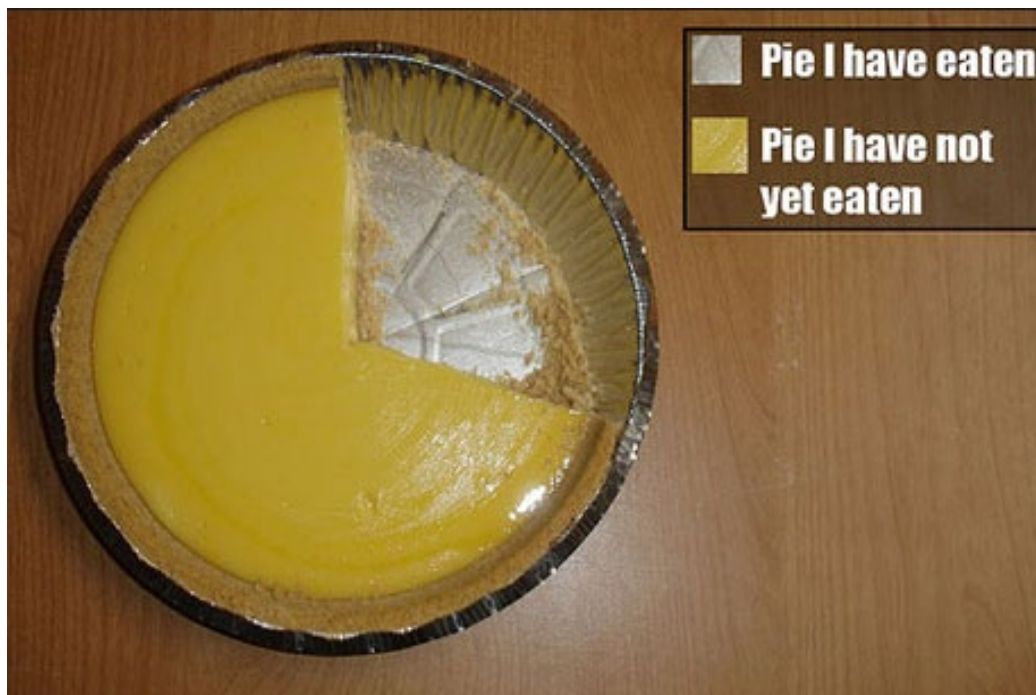
Five years since the end of the Great Recession, the economy has finally regained the nine million jobs it lost. But not all industries recovered equally. Each line below shows how the number of jobs has changed for a particular industry over the past decade. To see how the recession reshaped the nation's job market, click on any of the lines below.



<https://www.nytimes.com/interactive/2014/06/05/upshot/how-the-recession-reshaped-the-economy-in-255-charts.html>

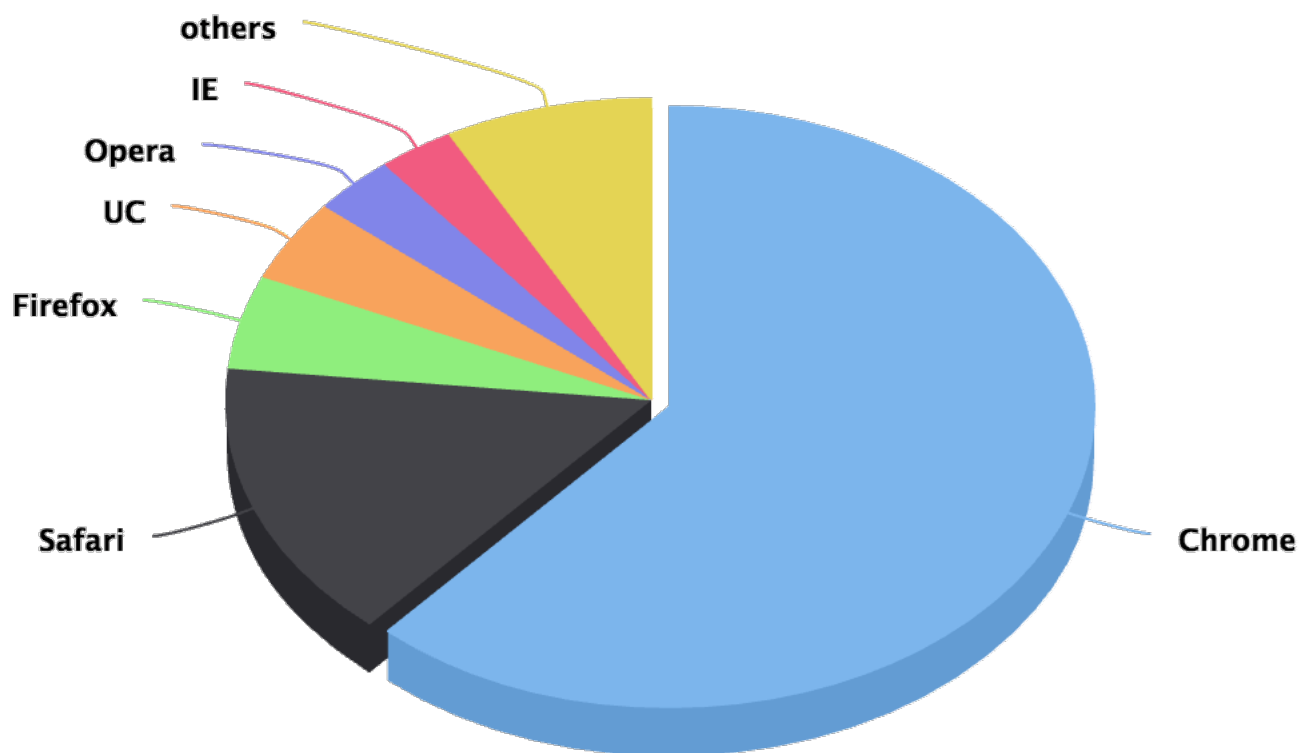
饼图

- 用角度展示比例



饼图

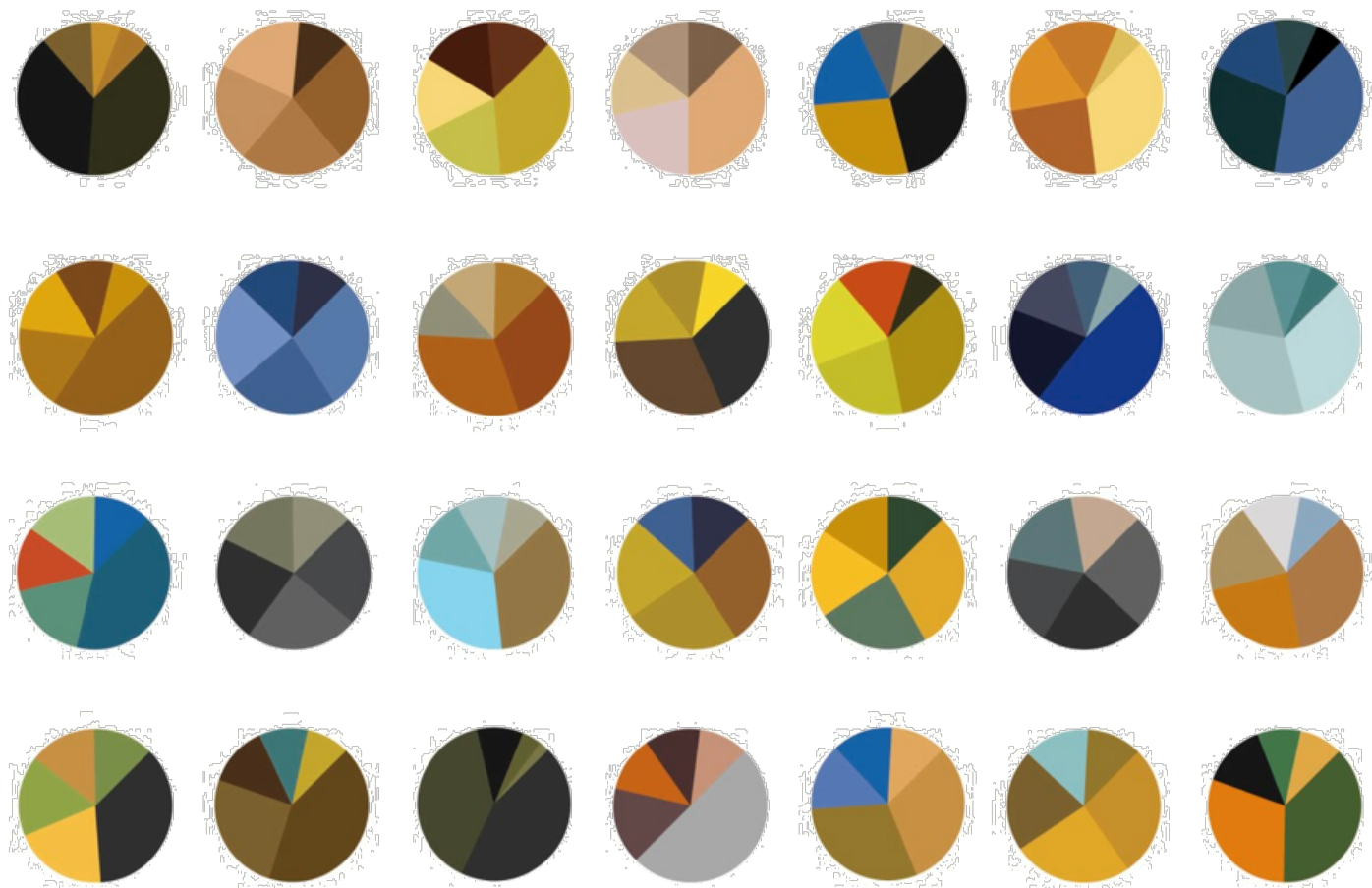
Browser market shares at a specific website, October 2018



饼图



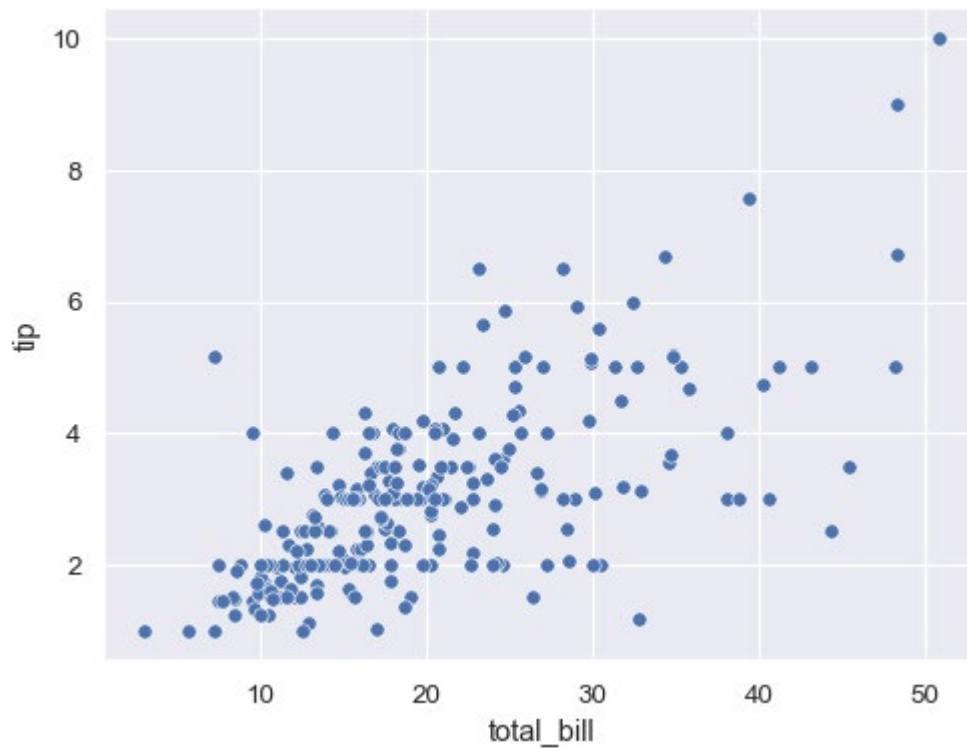
饼图



<https://www.arthurbuxton.com/#/art-design/>

散点图

- 通过散点的位置展示数据的分布



辅助线的作用

Pay Gap Between Women and Men

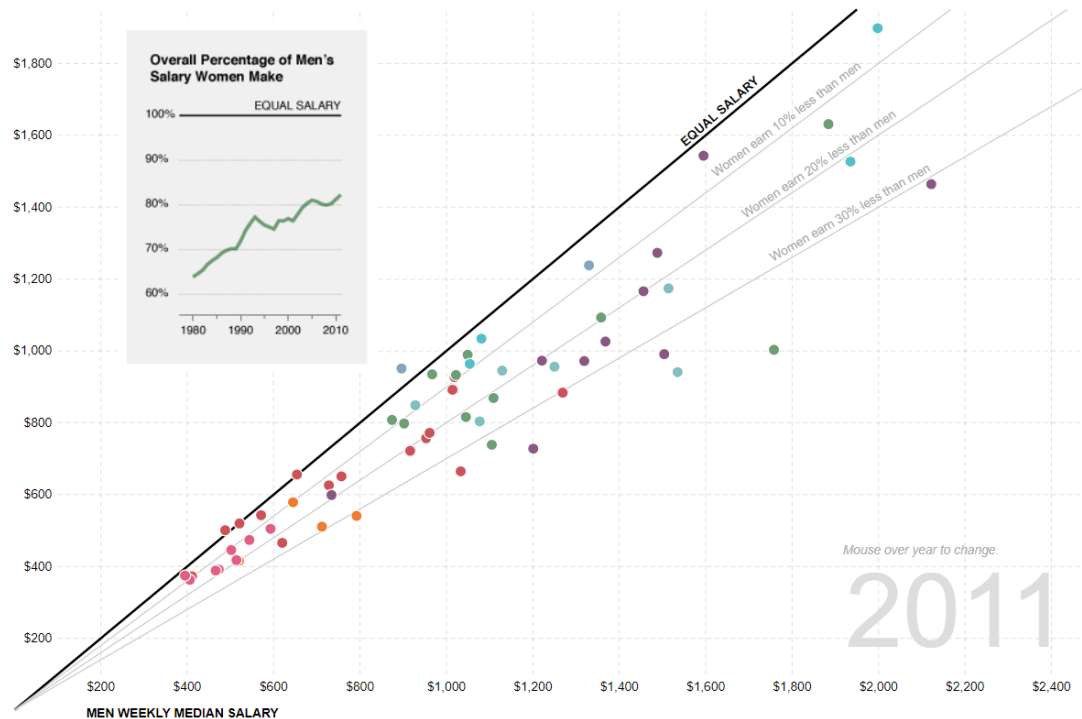
An update to [an interactive](#) by Hannah Fairfield and Graham Roberts of The New York Times in 2010, making use of Mike Bostock's [Wealth & Health of Nations D3 port](#).

On average, women are still paid less than men for working comparable jobs. Is it getting better? Below shows how salaries between the genders have changed over the past nine years.

Management	Computers & Mathematics	Service
Business Operations	Professional & Related	Production & Transportation
Sales & Office	Healthcare	

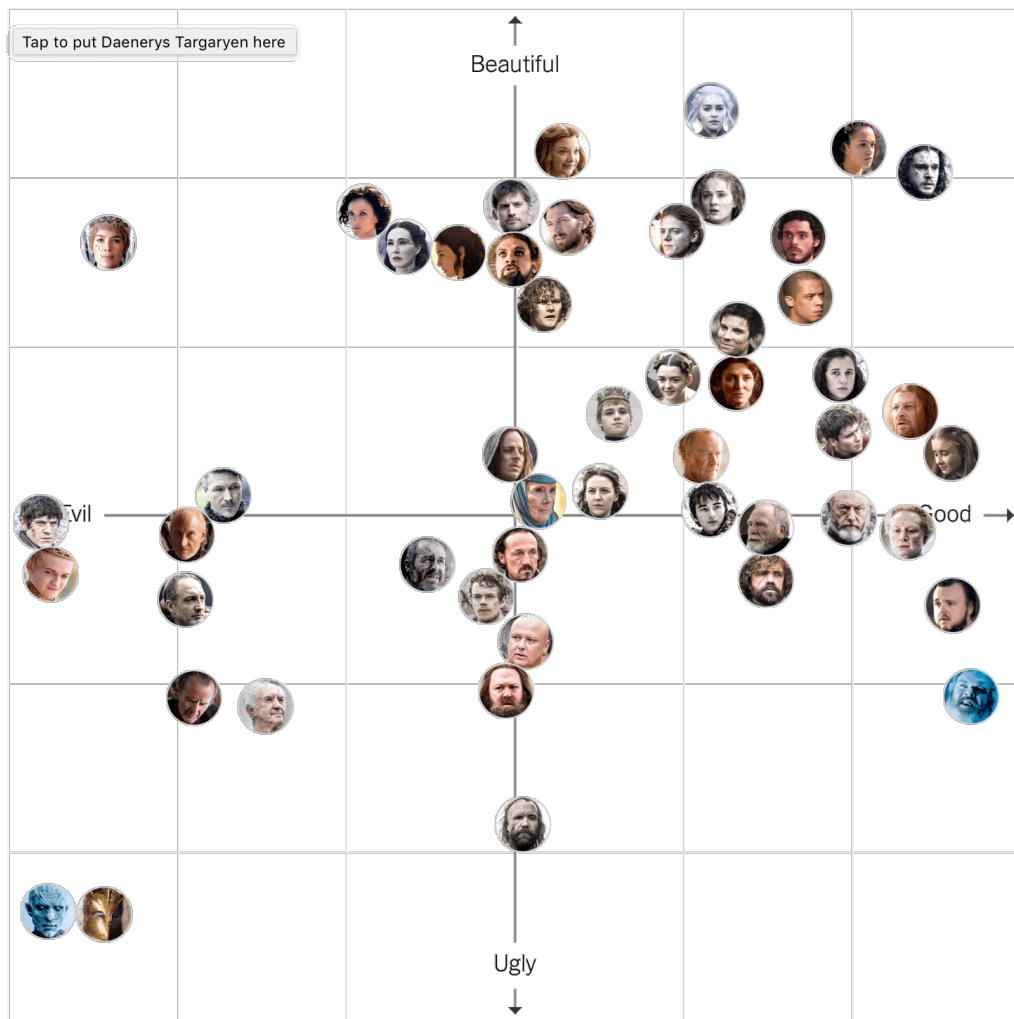
Show Paths

WOMEN WEEKLY MEDIAN SALARY



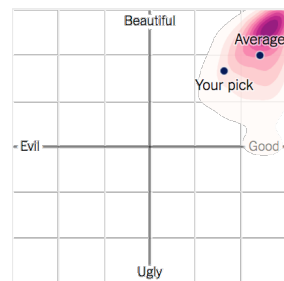
*Only occupations with data for all years and with at least 50,000 respondents for each sex are shown.
Source: Bureau of Labor Statistics — By [Nathan Yau](#)

散点图



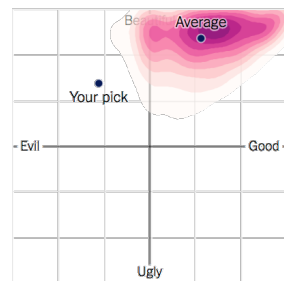
Jon Snow

The hero of the tale — so far. He's been the picture of fair and just and strong, though perhaps a little brooding at times.

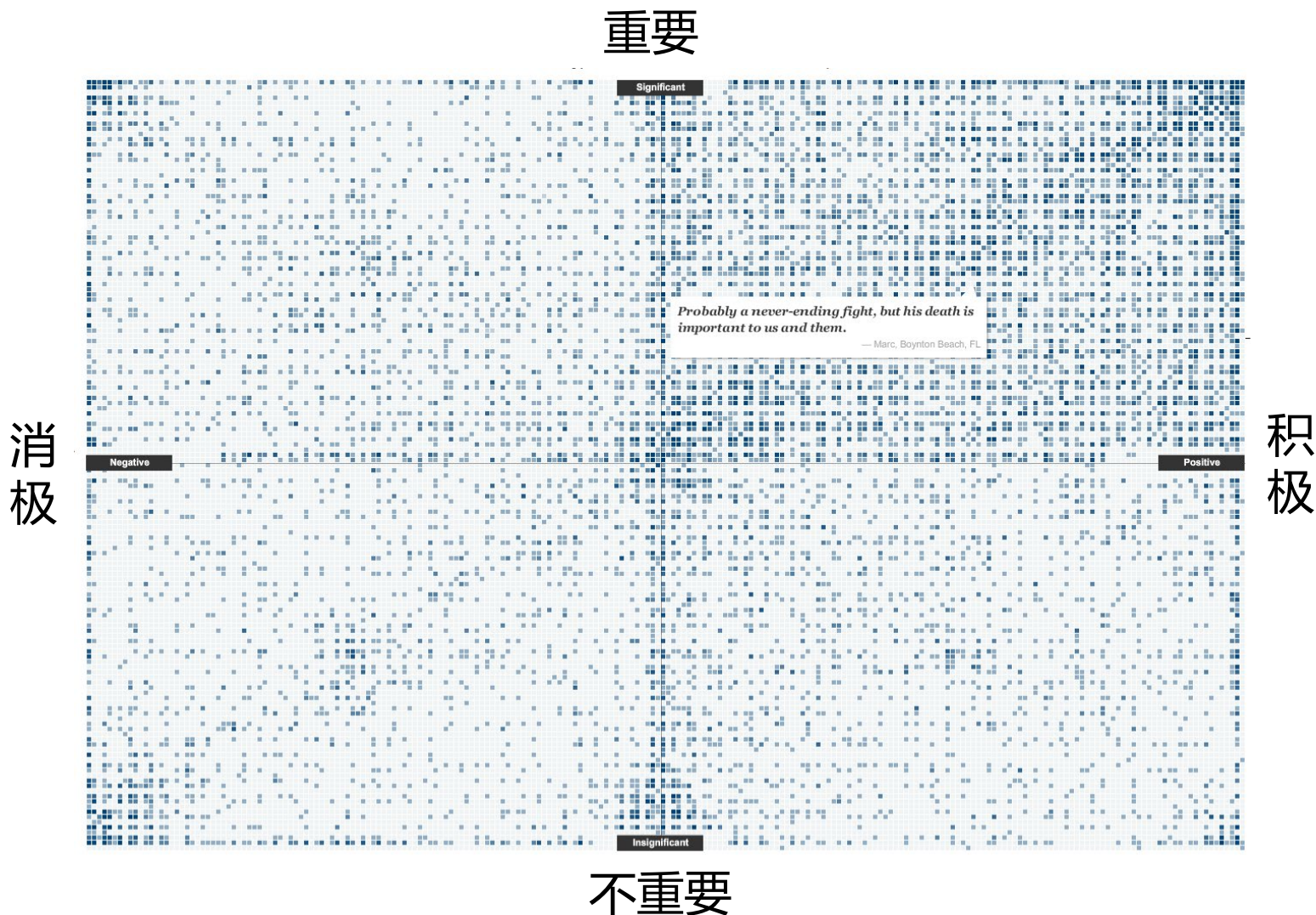


Daenerys Targaryen

A stunning beauty who feels for the oppressed. But she has no problem sending people to their deaths as she conquers kingdoms.

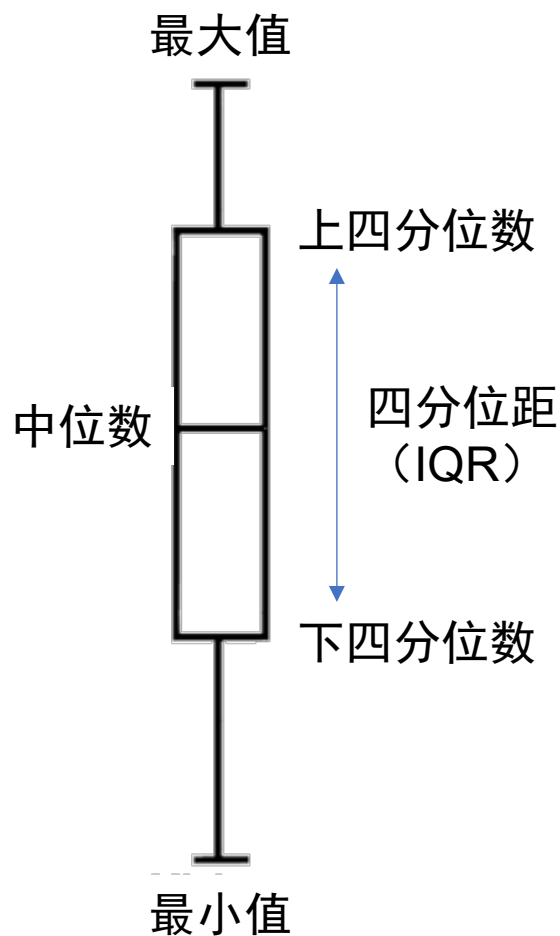


散点图变种

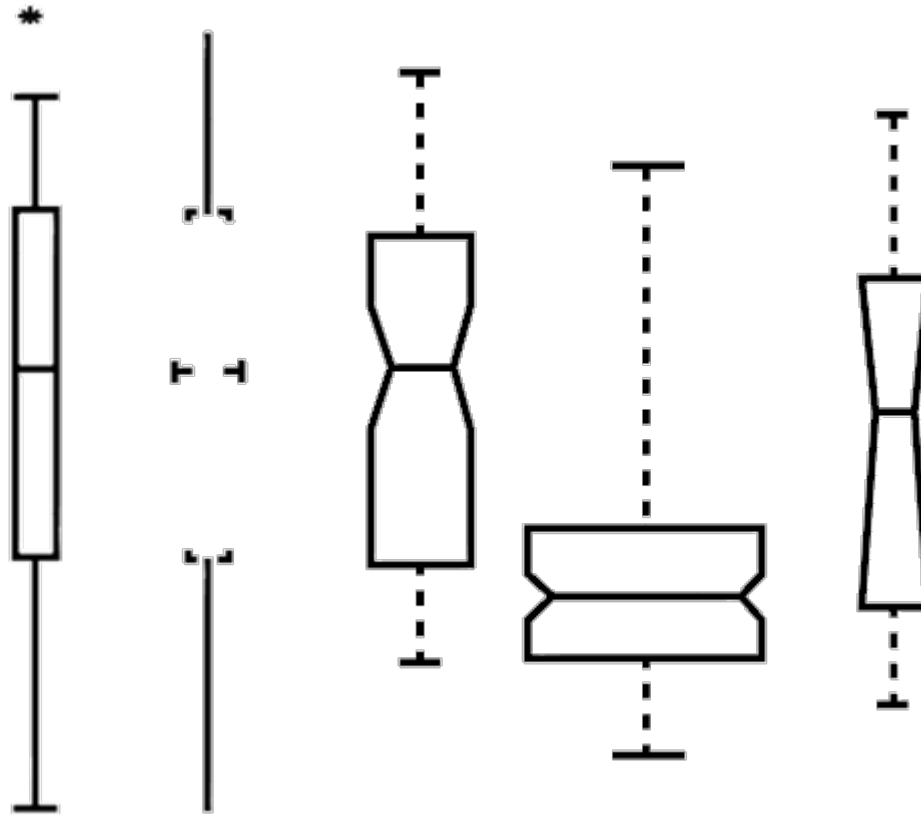


盒须图

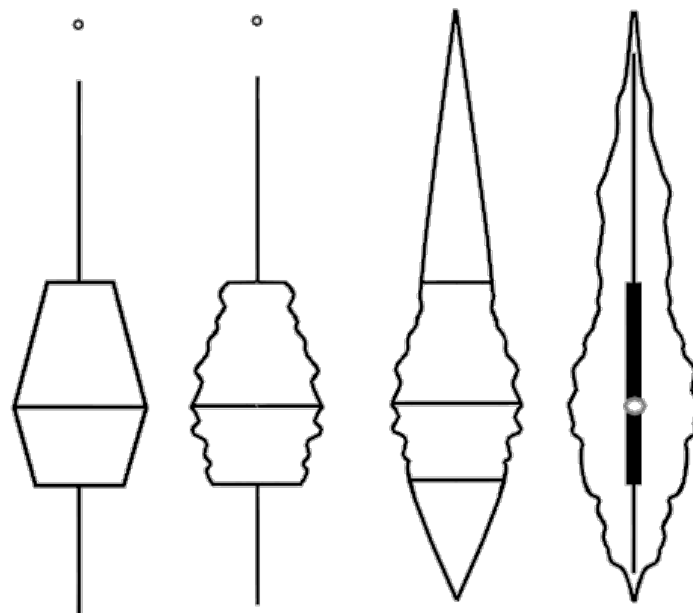
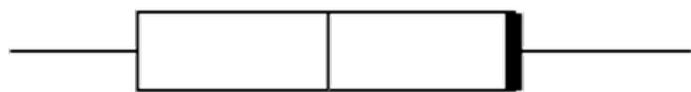
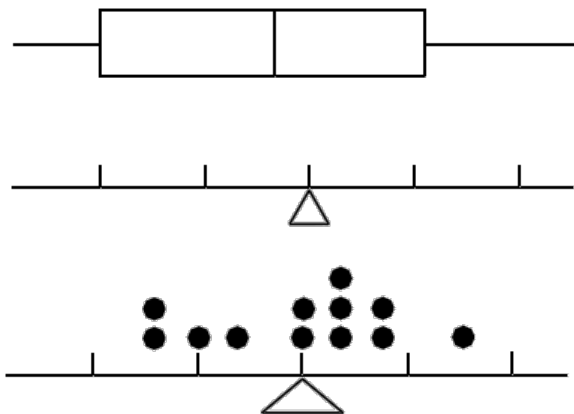
- 展示数据分布的重要分位点



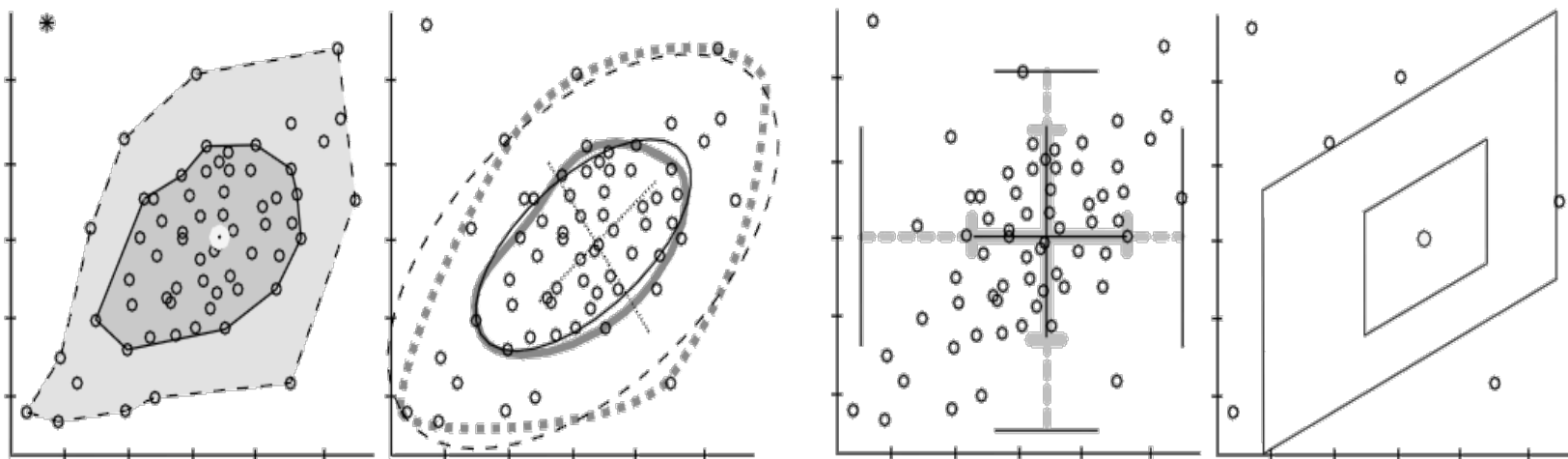
盒须图



盒须图的变种



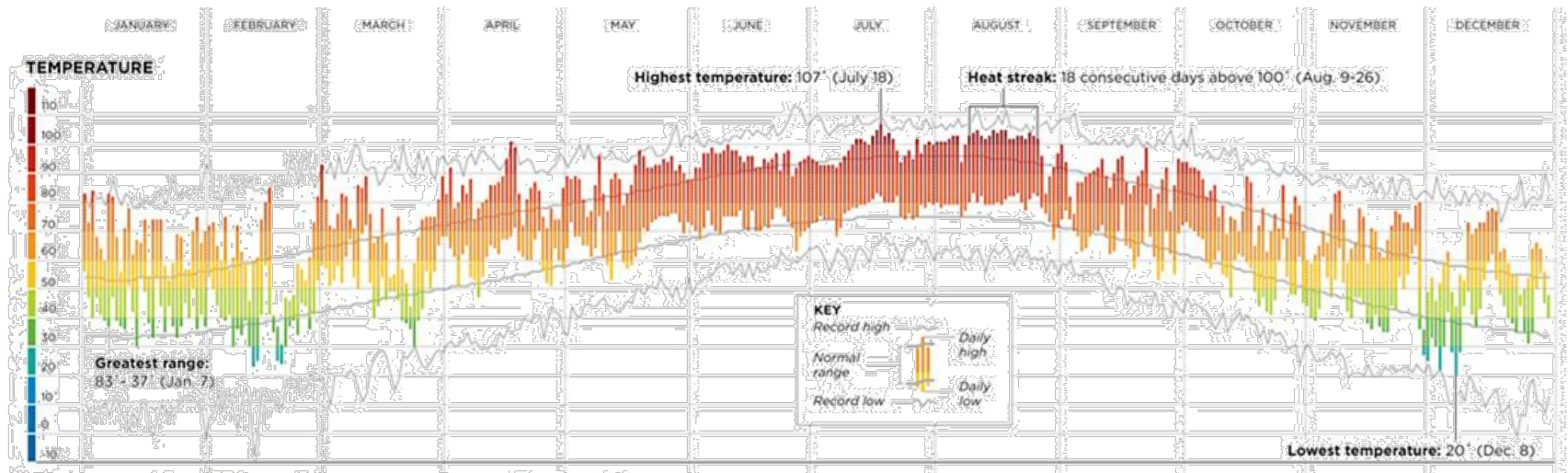
盒须图的变种



盒须图的变种

2006: The warmest year on record in the D-FW area

Drought persists as the annual average temperature increased for the fifth consecutive year with 43 days at 100° or above



针对数据的图表优化

- 数据分布严重失衡
- 数据量过大

分布失衡

THE WALL STREET JOURNAL.

□ DOW JONES | *Market Gaps*

FRIDAY, MARCH 27, 2020 - VOL. CCLXXV NO. 72

WSJ.com

★ ★ \$4.00

DIA 22552.17 ▲ 135162 6.4%

NASDAQ 7797.54 ▲ 5.8%

STOCK 600 321.30 ▲ 2.6%

10-YR. TREAS. ▲ 15/32, yield 0.806%

OIL \$22.60 ▼ \$1.89

GOLD \$1,650.30 ▲ \$17.81

EUR \$1.1035 YEN 109.57

What's News

Business & Finance

A record 3.28 million workers applied for unemployment benefits last week as the coronavirus hit the U.S. economy, marking an abrupt end to the nation's decade-long run of job growth. **A1, A2**

◆ U.S. stocks soared as the government came closer to approving a \$2 trillion stimulus package. The Dow returned to bull market territory, gaining 6.4%. The S&P 500 and Nasdaq rose 6.2% and 5.6%, respectively. **A1**

◆ The relief package passed by the Senate would help stabilize the coronavirus-battered economy, but likely isn't enough to bring it back to health. **A2, A6**

◆ Musklin indicated that the U.S. would take stakes in airlines in exchange for billions in direct grants to the firms, according to people familiar with the matter. **A6**

◆ G-20 leaders said they would collectively spend more than \$5 trillion trying to insulate the global economy from the impact of the pandemic. **A3**

◆ ViacomCBS's parent firm is losing some of its ability to borrow at a time when its key revenue drivers have ground to a halt. **A1**

◆ The Trump administration is moving ahead with new curbs aimed at cutting off Huawei from one of the

U.S. Virus Cases, Jobless Rate Soar

Historic run of growth abruptly comes to a close as unemployment claims soar to a record

WASHINGTON—A record 3.28 million workers applied last week as the new coronavirus hit the U.S. economy, marking an abrupt end to the nation's historic, decade-long run of job growth.

By Eric Morath,
Jon Hilsenrath
and Sarah Chaney

The number of Americans filing for claims was nearly five times the previous record. The surge was for the week ended March 21 and could rise, Pennsylvania, Ohio and California were among 10 states reporting more than 100,000 claims, leaving unemployment systems overloaded.

Millions of U.S. businesses have announced layoffs or furloughs, as their cash flows dry up. Several state and local authorities have ordered nonessential businesses to close in response to the novel coronavirus pandemic, bringing the great American job machine to a sudden halt.

Until March, U.S. employers added jobs for a record 113 straight months, causing payrolls to grow by 22 million. In the process, millions of people—including low-wage hourly laborers, disabled peo-

ple, minorities, former inmates and others—found work. The unemployment rate, which was 3.5% in February, had been at levels not seen since the 1960s. Wages started to accelerate in the past two years after lagging during the early stages of the expansion that followed the 2007-09 recession.

The strong labor market kept the U.S. economy humming for a decade.

And then, in a matter of days, it stopped.

Millions of Americans, already fearful the new coronavirus could infect them or their families, now have two new worries: When will the job machine start again? And can they hold out until it does? Much will depend on how long the virus crisis lasts and whether federal and state unemployment insurance pro-

grams can adapt quickly to fill the immense gaps building in household cash flows until the virus recedes.

"We haven't seen this big of a free fall before," said Keith Hall, former director of the Congressional Budget Office and advisor to President George W. Bush. "Not even during the Depression...It's really like an instant Great Recession."

Please turn to page A2

Number of infections across country pass China's count; New York remains epicenter

The U.S. has overtaken China as the country with the most cases in the world of the novel coronavirus, marking a new milestone in the fight against the global pandemic.

By Talal Ansari,
Jennifer Calais
and Chun Han Wong

China had held the top spot since the deadly virus began spreading there in December.

There were 83,507 confirmed infections across America, according to data compiled by Johns Hopkins University. The first case in the U.S.—a man in Washington state who had recently traveled to Wuhan, China—was reported on Jan. 21. Within six weeks, the U.S. began to see a surge in new infections and the country's first deaths.

New York is the center of the pandemic in the U.S., with its 37,802 infections accounting for nearly half of all cases in the country and tens of thousands more than any other state.

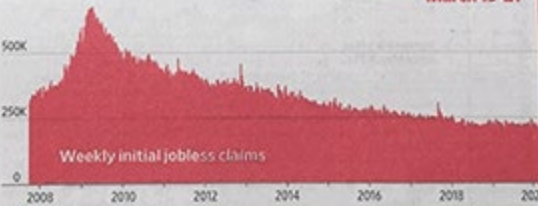
To help state and local authorities decide whether to bolster or relax social-distancing measures designed to help stop the spread of the virus, the Trump administration is planning to issue guidelines categorizing counties across

Please turn to page A4



National Guard troops in India, Calif., package food bags to assist people in home confinement.

3.28 million claims—
March 15-21



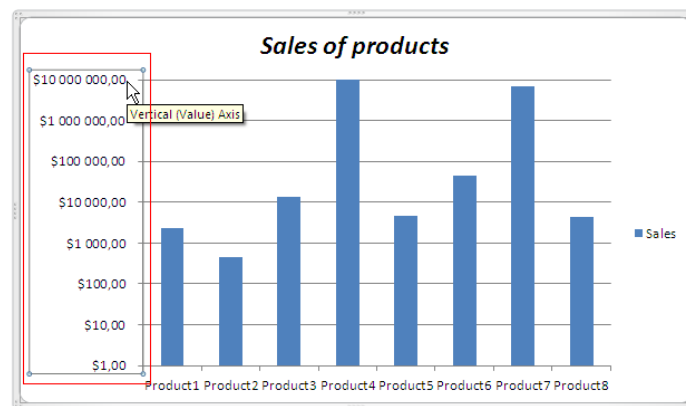
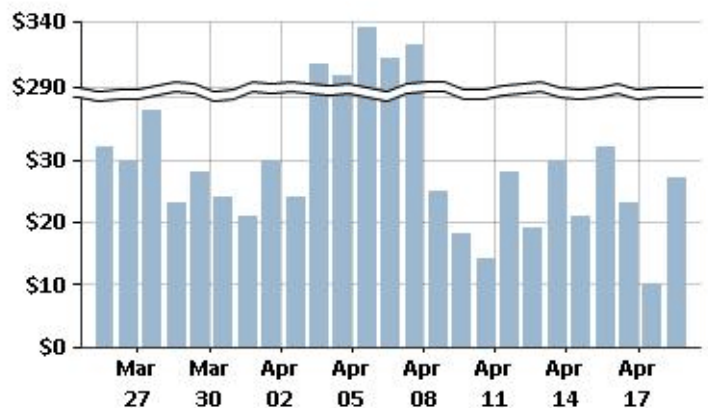
Note: Weekly claims are seasonally adjusted

Sources: Labor Department

◆ Rescue package to steady, not restore, economy. **A2**

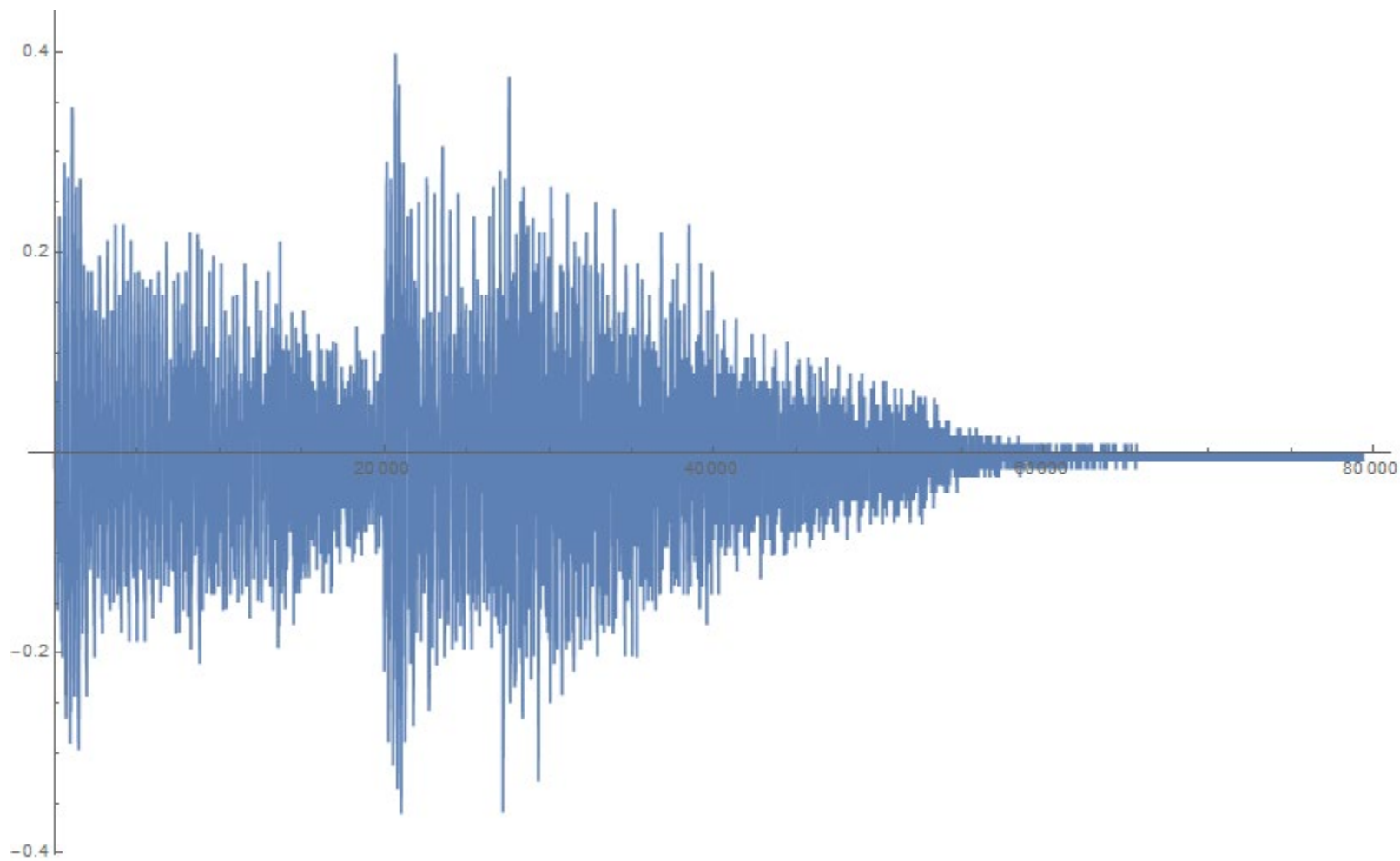
分布失衡

- 解决方法
 - 截断轴
 - 非线性映射（对数）



都会对比较造成干扰

数据量过大



总结

- 数据转换
 - 标准化/拟合/采样/离散化/聚类/降维
- 基础图表
 - 柱状图/折线图/饼图/散点图/盒须图